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**TANDON SCHOOL
OF ENGINEERING**

Data-Driven Material Selection for Fire-Safe Spacecraft Design

Final Report

Manef Elimami

New York University

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Abstract

Spacecraft fire safety requires material decisions under atmospheric, flow, geometric, and gravitational conditions for which direct testing is expensive and unevenly distributed. This report consolidates the project’s curated literature database and evaluates whether supervised machine learning can support two tasks: binary prediction of sustained ignition and regression of flame-spread rate (FSR). The evaluation deliberately separates interpolation among conditions represented within known papers from extrapolation to entire held-out papers. After deduplication, the ignition task contains 4,487 observations from 87 canonical papers; the FSR task contains 2,531 usable observations from 66 papers. Preprocessing, hyperparameter search, threshold selection, and augmentation are confined to training partitions.

For ignition, the selected row-level XGBoost model reaches a ROC–AUC of 0.9076 ± 0.0102 , whereas the selected unseen-paper XGBoost model reaches 0.7183 ± 0.0444 . For FSR, the selected random forest reaches $R^2 = 0.8790 \pm 0.0178$ and $\text{RMSE} = 0.0063 \pm 0.0005$ m/s for row-level interpolation; paper-grouped evaluation falls to $R^2 = 0.4576 \pm 0.2215$ and $\text{RMSE} = 0.0122 \pm 0.0041$ m/s. SHAP and permutation analyses identify oxygen, flow and transport quantities, thermal time scales, and fuel properties as important model inputs, but these associations are not interpreted causally. The central engineering result is therefore conditional: the database supports useful within-domain ranking and screening, but the generalization gap and source heterogeneity preclude using the models as stand-alone certification evidence. The models are positioned as traceable decision-support tools for prioritizing tests, identifying poorly covered regimes, and guiding database expansion.

Statement of scope and evidence

All numerical claims in this report are taken from version-controlled database validation files, frozen split metadata, model-selection records, and generated evaluation tables. The oral-defense deck is used to recover the intended narrative, not as the numerical source of record; it explicitly marks one metric page as not updated. Although the defense question mentioned extinction, completed pipelines exist only for ignition and FSR. Consequently, no extinction model or result is claimed here.

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Nomenclature

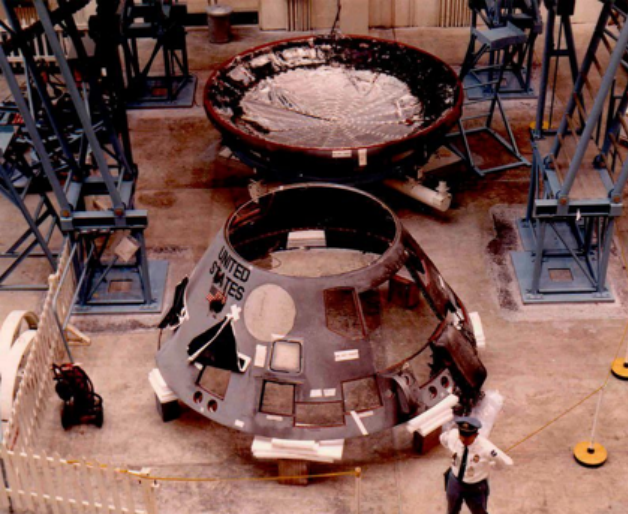
Symbol or acronym	Definition
c_p	Specific heat capacity, J/(kg K)
g	Acceleration level normalized by terrestrial gravity
k	Thermal conductivity, W/(m K)
α	Thermal diffusivity, m ² /s
ρ	Density, kg/m ³
FSR	Flame-spread rate, m/s
F1	Harmonic mean of precision and recall
FPR	False-positive rate
LOPO	Leave-one-paper-out
MAE	Mean absolute error
MCC	Matthews correlation coefficient
ML	Machine learning
MLP	Multilayer perceptron
OOD	Out of distribution
PR–AUC	Area under the precision–recall curve
RF	Random forest
RMSE	Root-mean-square error
ROC–AUC	Area under the receiver-operating-characteristic curve
SHAP	Shapley additive explanations
TPR	True-positive rate (recall, sensitivity)
XGBoost	Extreme gradient-boosted decision trees

1 Introduction

1.1 Spacecraft fire safety

Fire safety is a system-level requirement for crewed spacecraft. A pressurized vehicle necessarily combines oxidizer, combustible materials, and potential ignition sources in a confined volume where evacuation, replacement of damaged equipment, and atmospheric clean-up may be impossible. Electrical systems are indispensable to vehicle operation, yet faults, overheated components, and degraded insulation place ignition energy next to polymeric fuels. Material selection is consequently a compromise among fire performance, function, mass, manufacturability, and mission architecture rather than a simple elimination of all combustible items [1].

Apollo 1 remains the clearest demonstration of coupled fire hazards. During a ground test on 27 January 1967, a fire developed rapidly in the command module’s high-pressure pure-oxygen atmosphere. The exact ignition source was not established by the investigation, but electrical wiring was considered a likely source; extensive combustible material, the atmosphere, and the inward-opening hatch made the event catastrophic. The post-fire views in figure 1.1 show why specimen flammability cannot be separated from installation and vehicle configuration. The accident led to atmospheric, materials, wiring, hatch, and test-program changes, including extensive component and full-scale mock-up testing [1].



(a) Apollo 204 command module during the post-fire investigation.



(b) Prime crew: Grissom, White, and Chaffee.

Figure 1.1: Apollo 1 command module and crew. NASA public-domain photographs [4, 5].

Deep-space missions increase the consequences of such an event. A distant vehicle or surface habitat cannot assume rapid evacuation, immediate resupply, or prompt replacement of discharged extinguishers and damaged life-support hardware. Proposed low-pressure atmospheres, elevated oxygen fraction, lunar or Martian gravity, radiation aging, dust, waste accumulation, and long autonomous operation all interact with fire risk [1]. A material qualified in one terrestrial configuration therefore cannot be assumed to keep the same hazard ranking in every spacecraft environment.

1.2 Flammability, quenching, and blowoff

In this report, *flammability* is not reduced to a single binary property. It is represented by five complementary experimental observables:

These variables describe different stages and consequences of a fire. Ignition addresses initiation; flame length, FSR, and HRR characterize growth; smoke and aerosols affect detection and habitability. Their availability is uneven across the literature. The implemented machine-learning pipelines model only ignition and FSR. Flame length, HRR, and smoke/aerosols are retained as database outputs but excluded from predictors to prevent outcome leakage.

The stability of an established flame is bounded by two distinct extinction mechanisms. At high imposed flow or stretch, the gas residence time through the reaction zone becomes too short relative to the chemical reaction time. The flame detaches or is extinguished by *blowoff*, conventionally associated with a low Damköhler number. Self-induced buoyant flow can also produce blowoff during upward spread in normal gravity [2].

Table 1.1: Five variables used to describe material flammability in the database.

Variable		Engineering interpretation
Ignition		Whether the imposed energy and environment establish a flame.
Flame length		Spatial extent of the visible flame under the test condition.
Flame-spread (FSR)	rate	Speed of the advancing flame front, reported in m/s.
Heat-release (HRR)	rate	Rate of chemical energy release, reported in W when available.
Smoke/aerosols		Production of particulate and condensable combustion products relevant to visibility, toxicity, and detection.

At the opposite limit, very weak flow supplies insufficient convective heat feedback and oxygen transport to offset radiative and conductive losses. The flame shrinks and extinguishes by *quenching*. The low-speed quenching branch is experimentally accessible in microgravity because buoyancy no longer imposes a minimum natural-convection velocity. In ordinary gravity, buoyant flow masks this near-zero-flow branch; this is the precise sense in which the quenching limit considered here is observed only under microgravity conditions, not a claim that every possible form of flame quenching is exclusive to space.

Plotting limiting oxygen concentration against forced stretch or flow gives the U-shaped boundary in figure 1.2. The left branch is the microgravity low-flow quenching limit, the right branch is the high-flow blowoff limit, and stable burning occupies the region above the boundary. The minimum identifies the lowest oxygen concentration at which any tested flow can sustain combustion. This map demonstrates why oxygen concentration alone is insufficient: the same material and atmosphere may be non-flammable at very low and very high flow yet burn at an intermediate flow.

The remainder of this report reviews the spacecraft, combustion, and machine-learning literature that frames the research gap; states the resulting question and objectives; describes the database and evaluation design; and reports ignition and flame-spread predictions under interpolation and paper-held-out extrapolation.

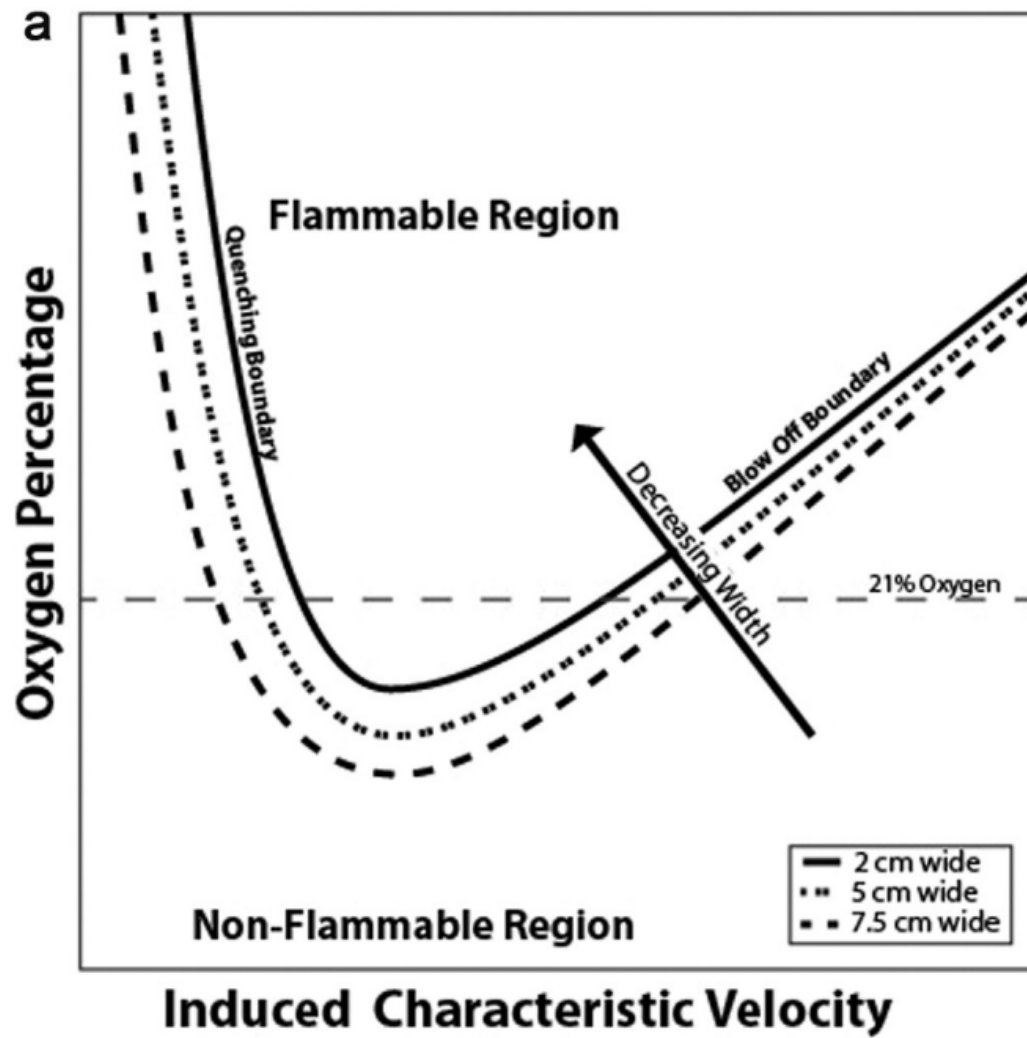


Figure 1.2: Schematic U-shaped flammability boundary. The low-velocity branch is quenching and the high-velocity branch is blowoff; curves show the reported fuel-width dependence. Reproduced from Johnston et al. [2].

2 Literature Review

2.1 Review strategy and source roles

We synthesize five sources that address different layers of the research problem. Guibaud et al. provide the spacecraft fire-safety context, historical evidence, and deep-space challenges [1]. Rivera et al. review the physics of flame spread over electrical wires and present the closest prior machine-learning study to the present FSR task [7]. Schwartz-Ziv and Armon provide comparative evidence about model selection for tabular data [9]. Raissi, Perdikaris, and Karniadakis establish the PINN formulation [6], while Karniadakis et al. situate physics-informed learning within the wider scientific-machine-learning field [3].

These source roles must remain distinct. Historical incident counts, spacecraft atmosphere descriptions, and deep-space concerns are attributed to Guibaud et al. Physical and machine-learning statements specific to wire flame spread are attributed to Rivera et al. Claims about tabular algorithm comparisons are limited to the datasets and protocol of Schwartz-Ziv and Armon. PINN equations describe methods reported in the PINN literature; they do not imply that this project’s tabular pipelines solved a combustion PDE. Finally, none of the performance values reported for earlier studies is presented as a result of this project.

Table 2.1: Authoritative sources and their use in this report.

Source	Principal evidence used	Limitation for the present question
Guibaud et al. [1]	Spacecraft incidents, fire-safety strategy, reduced-buoyancy implications, and deep-space challenges	Review and risk context; not a trained ignition or FSR predictor
Rivera et al. [7]	Wire-flame physics, heterogeneous experimental predictors, categorization, augmentation, and ML FSR benchmarks	Focuses on wire FSR; random cross-validation is not strict held-source-out transfer
Shwartz-Ziv and Armon [9]	Controlled comparison of XGBoost with deep models on eleven tabular datasets	General tabular evidence, not combustion-specific evidence
Raissi et al. [6]	Differential-equation residuals, collocation points, and forward/inverse PINN formulation	Assumes suitable governing equations and coordinate-resolved information
Karniadakis et al. [3]	Broader synthesis of physics-informed machine learning and data-physics integration	Research framework rather than a spacecraft-material qualification method

2.2 Spacecraft fire safety as a material-selection problem

2.2.1 Lessons from incidents and evolving practice

The history reviewed by Guibaud et al. shows that spacecraft fire safety evolved through incident investigation, atmospheric redesign, material screening, detection hardware, suppression equipment, and crew procedures. Their chronology identifies twelve acknowledged fire or smoke events. Apollo 1 remains the clearest demonstration of coupled design hazards: an oxygen-rich cabin, combustible interior materials, possible electrical ignition, rapid pressure rise, and an inward-opening hatch combined in a fatal accident during a ground test. The precise ignition source was not established, but subsequent actions included atmospheric changes, removal or protection of combustibles, and extensive component and mock-up flammability testing [1].

Apollo 1 also demonstrated that material behaviour cannot always be inferred from an isolated specimen. Guibaud et al. report that components showing acceptable individual behaviour did not necessarily prevent excessive propagation after integration into a full spacecraft mock-up. The configuration changed the available fuel paths, heat feedback, and interactions among adjacent items. This observation remains relevant to data-driven screening: a row-level material prediction can inform a decision, but cannot by itself represent every installation detail in a complete vehicle [1].

Later incidents shifted attention toward electrical equipment and operational response. The Salyut and Shuttle records include failed fans, short circuits, overheated resistors, fused wires, and failed electronic components. In the five Shuttle events summarized by Guibaud et al., the crew interrupted power before a full-scale fire developed, while onboard detection systems did not provide the decisive first warning. The 1997 Mir fire showed a different failure mode: a lithium-perchlorate oxygen canister produced an energetic flame and dense smoke, obstructed access to one escape vehicle, and could not be directly suppressed by the available water-based foam. Crew members had

to improvise while contending with visibility, equipment access, and respirator problems [1]. The literature’s point is not that every spacecraft fire resembles a wire fire, but that ignition source, material, atmosphere, access, and mitigation interact at system scale.

Present practice uses multiple defence layers. Guibaud et al. describe fire-resistant material selection, configuration analysis, redundant or distributed smoke detection, electrical and ventilation shutdown, portable extinguishers, atmospheric monitoring, crew masks, and rehearsed procedures on the International Space Station. This layered architecture has benefited from earlier programmes, yet it still relies on knowledge of material response and combustion signatures in low gravity. The review argues that uncertainties remain in translating terrestrial test performance to reduced gravity and in demonstrating suppression performance for every relevant scenario [1].

2.2.2 Qualification tests and their limits

Spacecraft material control is intentionally conservative. The test history summarized by Guibaud et al. includes NASA’s vertical flame-propagation screening, in which multiple specimens are exposed to an ignition source under a specified atmosphere. The cited criterion considers both propagation distance and ignition by burning debris. At the vehicle level, a material that does not pass an isolated test may still require configuration analysis when mission-critical function prevents simple removal. Separation, containment, barriers, and elimination of propagation paths become part of the safety argument [1].

This approach is essential but does not produce a continuous map over all operating conditions. A pass/fail test answers a deliberately bounded qualification question. It does not directly estimate a probability of ignition for every pressure and oxygen concentration, nor does it yield FSR over arbitrary flow and gravity. Furthermore, Guibaud et al. cite reduced-gravity observations in which limiting behaviour differs from normal-gravity expectations and describe partial-gravity studies where relative material ranking can change [1]. A data-driven surrogate is useful precisely in this space between discrete experiments: it can organize existing observations, interpolate under controlled assumptions, and expose regions where direct testing remains necessary. It cannot supersede the qualification test from which the safety requirement is defined.

2.3 Ignition and flame spread under reduced buoyancy

2.3.1 Coupled solid- and gas-phase processes

Ignition is a transition rather than an intrinsic label attached permanently to a material. Energy supplied by an igniter or fault heats the solid; heat is stored and conducted through the specimen and, if the material decomposes, volatile fuel is generated. Gas-phase transport then mixes fuel and oxidizer, while losses to the core, surroundings, and unheated material compete with heat feedback. The literature reviewed by Rivera et al. emphasizes that even reduced analytical descriptions require material properties, dimensions, gas properties, and heat-flux estimates that are not uniformly reported across experiments [7]. Consequently, an observed yes/no ignition outcome is conditional on the entire experimental configuration.

A useful scale for solid heating is the thermal diffusion time

$$t_{\text{diff}} \sim \frac{\ell_c^2}{\alpha_s}, \quad \alpha_s = \frac{k_s}{\rho_s c_{p,s}}, \quad (2.1)$$

where ℓ_c is a characteristic solid length, k_s is thermal conductivity in $\text{W m}^{-1} \text{K}^{-1}$, ρ_s is density in kg m^{-3} , $c_{p,s}$ is specific heat in $\text{J kg}^{-1} \text{K}^{-1}$, and α_s is thermal diffusivity in $\text{m}^2 \text{s}^{-1}$. This scaling does not determine ignition by itself. It shows why thickness and thermophysical properties jointly affect the time over which a specimen responds to heating, and why separately recorded database columns can have interacting rather than additive effects.

After ignition, flame spread requires successive heating and pyrolysis of virgin material. Rivera et al. reproduce an analytical opposed-flow flame-spread relation in which spread speed depends on heated length, solid thermal inertia, pyrolysis and ambient temperatures, several flame and external heat-flux terms, radiative loss, and a kinetic contribution [7]. At the level needed here, that balance can be written schematically as

$$v_{\text{FSR}} \sim \frac{\dot{q}_{\text{net}}'' \ell_h}{\rho_s c_{p,s} \delta (T_p - T_0)}, \quad (2.2)$$

where $\dot{q}_{\text{net}}''$ denotes net heating of the unburned solid, ℓ_h is a heated length, δ is a solid thickness scale, T_p is a pyrolysis-temperature scale, and T_0 is the initial temperature. Equation (2.2) is a qualitative energy-balance representation, not the regression equation used by this project. It makes the key coupling explicit: any variable that changes heat feedback, heat storage, or heat loss can alter spread speed.

2.3.2 Microgravity transport

On Earth, gravity-induced buoyancy contributes a characteristic velocity and organizes the flame and smoke plume. In microgravity that contribution is suppressed, so imposed ventilation and diffusion control transport to a much

greater degree. Guibaud et al. connect this change to flammability, spread, smoke movement, detector response, and suppression uncertainty. A fire may produce combustion products that take longer to reach a sensor, while a relatively small ventilation flow can reshape a non-buoyant flame [1]. This is why “zero gravity” cannot be represented only as one more categorical label while all other transport variables are ignored.

Dimensionless groups help organize the competing mechanisms. For characteristic velocity U , length L , gas kinematic viscosity ν , and gas thermal diffusivity α_g ,

$$\text{Re} = \frac{UL}{\nu}, \quad \text{Pe} = \frac{UL}{\alpha_g}, \quad \text{Pr} = \frac{\nu}{\alpha_g}. \quad (2.3)$$

These groups compare inertial to viscous transport, advection to thermal diffusion, and momentum to thermal diffusivity, respectively. Their value in a heterogeneous database is representational: experiments with different dimensions and velocities may occupy related transport regimes. They do not remove the need to preserve dimensional predictors or account for geometry.

Opposed and concurrent flow are physically different configurations. In opposed spread, gas flow is directed toward the advancing flame front; in concurrent spread, flow and flame advance share a direction. Rivera et al. identify spread direction as an important categorical descriptor and report that adding such regime information substantially improved predictions in their compiled wire dataset [7]. The result supports explicit representation of configuration, but it must be interpreted within that study’s cross-validation protocol. It does not prove that a direction label captures all flow physics.

Partial gravity adds further complexity. Guibaud et al. review experiments and modelling that indicate non-monotonic responses for some fire quantities between microgravity and terrestrial gravity. They report different trends for upward and downward propagation and note that material ranking may change between normal, lunar, and Martian gravity [1]. The engineering consequence is that a predictor should not silently infer partial-gravity behaviour from an arbitrary linear interpolation between microgravity and terrestrial gravity. Sparse coverage must instead be treated as an extrapolation problem and reported as such.

2.3.3 Atmosphere, pressure, and oxygen

Oxygen concentration and pressure appear repeatedly in the spacecraft literature because mission atmosphere is both a life-support and structural design choice. Guibaud et al. describe the pure-oxygen, low-pressure atmosphere used in early American flight and the much higher pressure pure-oxygen condition present during the Apollo 1 ground test. Later spacecraft generally adopted nitrogen-diluted atmospheres, while proposed exploration vehicles may again use reduced total pressure and elevated oxygen fraction to address structural mass and extravehicular-activity considerations [1]. Fire response must therefore be considered over a domain, not only at terrestrial sea-level conditions.

Oxygen fraction is not equivalent to oxygen partial pressure, and neither alone captures all pressure effects. For an ideal mixture the oxygen partial pressure is

$$p_{\text{O}_2} = X_{\text{O}_2} P, \quad (2.4)$$

with mole fraction X_{O_2} and total pressure P in pascals. Equal values of p_{O_2} can still coexist with different total gas density and transport properties when diluent and total pressure differ. Rivera et al. consequently include oxygen concentration and pressure as separate predictors in the wire FSR problem [7]. This supports a multivariate representation rather than collapse to a single atmosphere variable.

2.4 Electrical-wire flame spread

Electrical wires are a particularly relevant material system because conductor and insulation form a coupled thermal geometry. Rivera et al. synthesize experiments using laboratory wires with metal cores and polymer insulation. Their review identifies core material and diameter, insulation material and thickness, oxygen concentration, pressure, forced flow, gravity, and orientation as influential quantities [7]. These variables also illustrate why the broader database is heterogeneous: a nominal material may appear in different specimen forms, and the same environment may produce different outcomes for different length scales.

The central conductor has a dual role. A high-conductivity core can move heat ahead of the flame and assist preheating. The same core can drain energy away from the reaction and pyrolysis region, especially when its cross-section is large relative to the insulation. Rivera et al. attribute this balance to core and insulation dimensions and properties, environmental conditions, and gravity [7]. A monotonic rule such as “higher core conductivity always increases FSR” would therefore contradict the reviewed physics. Flexible nonlinear models are attractive because they can represent conditional interactions, but flexibility also increases the need for careful validation.

Insulation behaviour introduces discrete regimes. Polymer type affects thermophysical response, while melting and dripping can change fuel geometry and energy transport. Orientation determines whether gravity assists movement of molten insulation in normal or partial gravity. Rivera et al. therefore encode core material, insulation material, spread direction, and configuration as categorical predictors in addition to numerical properties [7]. This

is an important methodological lesson: a table containing only bulk conductivity, density, and heat capacity may omit morphological or chemical distinctions that those properties do not fully identify.

The wire literature also exposes a transferability issue. Studies may concentrate on one polymer, core, diameter range, apparatus, pressure regime, or flow direction. When observations from the same study are divided randomly among training and test folds, the test set can remain close to the training distribution. Rivera et al. acknowledge that their random ten-fold cross-validation measures robustness within the compiled literature dataset rather than strict transfer to an entirely new source [7]. For material selection, this distinction is critical because a new candidate is often associated with a new material family or experimental campaign.

2.5 Machine learning for flame-spread prediction

2.5.1 Closest prior work

Rivera et al. provide the closest direct precedent for data-driven FSR prediction. They compile approximately 1200 wire-flame observations and describe numerical features for core, insulation, geometry, atmosphere, flow, and gravity, together with categorical indicators for material and spread configuration. They compare linear regression, decision tree, k -nearest neighbours, elastic net, stochastic gradient descent, Bayesian ridge, gradient boosting, support-vector regression, kernel ridge, and a multilayer perceptron [7]. The breadth of this benchmark is useful because it avoids treating one model family as the default.

Their baseline using numerical predictors alone produced weaker performance than versions that included one-hot encoded categorical variables. Under categorized random ten-fold cross-validation, several nonlinear models attained mean coefficients of determination around or above 0.90; the exact values vary by model and augmentation strategy [7]. These are literature results. They cannot be transferred directly to this project’s database, targets, partitions, or code, and they do not establish held-paper generalization.

Rivera et al. also compare bootstrap resampling, small numerical perturbations, and a generative adversarial network (GAN). Bootstrap augmentation was the most consistently beneficial for tree- and neighbourhood-based models in their reported experiments. The authors stress that resampling preserves observed combinations, whereas noise and GAN generation can produce statistically plausible combinations without guaranteeing geometric consistency, conservation, or other physical constraints [7]. This qualification matters: synthetic tabular rows are not new experiments. If material density, conductivity, dimensions, and categorical identity are perturbed independently, the resulting row may not correspond to a realizable specimen.

The prior study reports a small number of negative FSR predictions from unconstrained regressors and correctly identifies them as physically inadmissible [7]. This motivates non-negative output handling, target transformation, or explicit post-fit diagnostics. More generally, machine-learning performance must be judged against physical admissibility as well as aggregate error. A model can obtain a favourable mean score while failing in the low-FSR region that is most relevant to a safety boundary.

2.5.2 Model choice for tabular data

The structure of the combustion database resembles the broad statistical category studied by Shwartz-Ziv and Armon: fixed columns, heterogeneous units, mixed numerical and categorical features, missingness, and no spatial or sequential locality comparable to pixels or words. Their controlled study evaluates four deep tabular architectures and XGBoost across eleven classification and regression datasets, with a common hyperparameter-optimization protocol. XGBoost outperformed the deep models on most of the considered datasets, particularly those not used in the original deep-model papers, and its tuning converged more readily in their experiments [9].

The conclusion is narrower than “deep learning never works for tables.” Shwartz-Ziv and Armon explicitly invoke the no-free-lunch principle and report that an ensemble containing both deep models and XGBoost achieved the best average relative performance in their comparison [9]. For this project, the defensible inference is to include gradient-boosted trees as a strong baseline and compare them fairly with other candidates. Model selection must follow held-out evidence under the relevant engineering split, not reputation or architectural novelty.

Tree ensembles are plausible for this task because thresholded partitions can capture nonlinear interactions and accommodate regime changes, while neural and kernel models offer different smoothness and representation biases. No family removes the need for leakage control. Imputation, scaling, categorical encoding, feature selection, and hyperparameter optimization must be fitted using training data only. Repeated measurements and paper-specific clusters should remain grouped where the evaluation question concerns transfer to a new source. These are methodological requirements of the present project, not performance claims borrowed from the cited tabular benchmark.

2.6 Physics-informed machine learning

2.6.1 PINN formulation

Raissi et al. define PINNs for systems represented by nonlinear differential equations. For a latent field $u(t, \mathbf{x})$,

$$\frac{\partial u}{\partial t} + \mathcal{N}[u; \boldsymbol{\lambda}] = 0, \quad \mathbf{x} \in \Omega, \quad t \in [0, T], \quad (2.5)$$

the neural network approximates u , and automatic differentiation evaluates the residual

$$r_\theta(t, \mathbf{x}) = \frac{\partial u_\theta}{\partial t} + \mathcal{N}[u_\theta; \boldsymbol{\lambda}]. \quad (2.6)$$

Training combines mismatch to available observations and initial or boundary data with residual error at collocation points,

$$\mathcal{L}(\theta) = \frac{1}{N_u} \sum_{i=1}^{N_u} |u_\theta(t_i, \mathbf{x}_i) - u_i|^2 + \frac{1}{N_r} \sum_{j=1}^{N_r} |r_\theta(t_j, \mathbf{x}_j)|^2. \quad (2.7)$$

The framework supports forward inference with known physical parameters and inverse identification in which parameters in $\mathcal{N}$ are learned from observations [6].

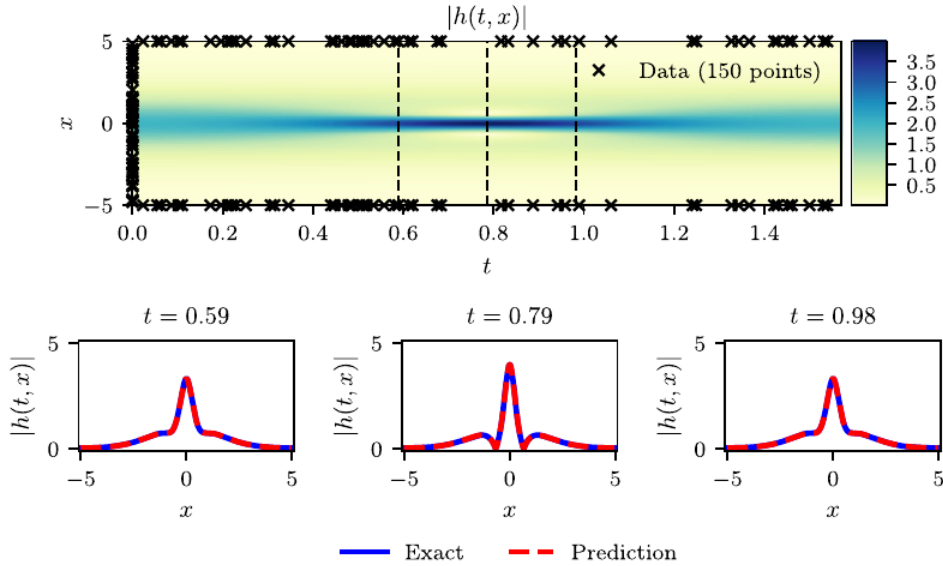


Figure 2.1: Example physics-informed neural-network solution from Raissi et al.: sparse initial and boundary observations and interior collocation points constrain the nonlinear Schrödinger solution; predicted profiles (red dashed) are compared with the exact solution (blue). Reproduced from Fig. 1 of [6].

Raissi et al. demonstrate the formulation on nonlinear Schrödinger, Allen–Cahn, Navier–Stokes, Korteweg–de Vries, and Burgers equations, using continuous- and discrete-time variants. They emphasize that PINNs complement rather than replace mature numerical methods and identify unresolved issues involving architecture, optimization, convergence, noise, and uncertainty [6]. Karniadakis et al. broaden the discussion to scientific settings where observational data may be scarce, noisy, or incomplete and prior physical knowledge can constrain an otherwise large hypothesis space [3].

2.6.2 Relevance and limitation for combustion tables

The appeal to spacecraft combustion is evident. Reduced-gravity experiments are costly, and conservation or transport relations contain information not present in a small set of labels. A valid physics-informed constraint could discourage impossible predictions and improve data efficiency. Dimensionless groups, positivity, energy-balance trends, and known symmetries are possible levels of physical structure, ranging from feature engineering to hard or soft constraints. The PINN literature provides a principled reason to investigate such structure [3, 6].

However, the standard PINN formulation requires more than a list of experiment-level rows. It needs a specified state field, spatial and temporal coordinates, a governing differential operator, and sufficient initial or boundary information for the target problem. The curated database records outcomes such as ignition and FSR under varied apparatuses; it is not a common mesh of temperature, species concentration, velocity, and reaction-rate fields.

Applying Equation (2.7) without a defensible operator and boundary conditions would create the appearance of physics without its mathematical content.

This limitation is reinforced by Rivera et al., who conclude that the theoretical framework for their heterogeneous wire data remains incomplete for direct use of fully physics-informed models [7]. The scientifically appropriate hierarchy is therefore: preserve SI units and physically meaningful variables; encode known regimes; enforce indisputable output constraints such as non-negative FSR; evaluate transfer across sources; and reserve PDE-residual PINNs for datasets and subproblems where the governing fields and boundaries are actually available. This project can be physics-aware without claiming a PINN where none was implemented.

2.7 Synthesis, research gap, and objectives

The literature converges on four findings. First, spacecraft fire is a credible system hazard whose consequence increases when evacuation, resupply, and recovery are restricted [1]. Second, ignition and spread depend jointly on material, geometry, atmosphere, flow, and gravity; reduced buoyancy changes transport rather than merely slowing or accelerating a terrestrial flame by a fixed factor [1, 7]. Third, nonlinear supervised models can recover useful relationships from heterogeneous wire-flame tables, especially when physically meaningful categories are retained [7]. Fourth, strong tabular baselines and physical constraints should be evaluated empirically and applied only within the assumptions their source literature supports [3, 6, 9].

2.7.1 Research gap and data-driven motivation

Reduced gravity changes the balance among diffusion, imposed ventilation, radiation, and solid conduction. Experiments are correspondingly expensive, and the published evidence is fragmented across materials, geometries, facilities, atmospheres, and reporting conventions. A finite qualification matrix cannot dense-sample this space, while a high-fidelity reacting-flow model requires boundary conditions and material properties that are often absent from historical reports.

The research gap is therefore not simply a lack of another predictive algorithm. Three specific absences remain. The *target gap* is that the closest machine-learning study concentrates on FSR, whereas an engineering screening workflow also needs an explicit ignition model. The *validation gap* is that randomly splitting rows can place nearly repeated observations from one rig on both sides of the evaluation boundary, creating optimistic evidence for a material-selection question that usually concerns a new campaign. Random cross-validation within a compiled literature table does not measure transfer to a held-out paper, apparatus, or condition regime. The *integration gap* is that broad spacecraft studies establish operational relevance while wire and tabular-learning studies establish local predictive feasibility, but the evidence has not yet been organized into one traceable, leakage-resistant evaluation of both ignition and FSR.

A data-driven response is motivated by this fragmentation rather than by a claim that combustion physics is obsolete. What is missing is a unified database and an evaluation that asks both whether models interpolate among conditions from known campaigns and whether they transfer to a completely held-out paper. That distinction is the difference between recovering nearby points in a familiar apparatus and supporting a screening decision for a new material, geometry, or laboratory.

2.7.2 Research question and objectives

To what extent can supervised models trained on a curated, multi-source microgravity-combustion database predict (i) whether ignition occurs and (ii) the flame-spread rate under spacecraft-relevant material, geometric, atmospheric, flow, and gravity conditions, while maintaining performance when entire experimental papers are withheld?

The work has four objectives:

1. assemble a source-traceable database with consistent variables and SI units from heterogeneous publications;
2. construct clear, reproducible classification and regression pipelines;
3. compare row-level interpolation with paper-grouped extrapolation; and
4. interpret model behavior using held-out diagnostics, permutation importance, and SHAP without treating predictive associations as causal laws.

The intended role is screening and experiment prioritization, not replacement of qualification tests or combustion physics. Only ignition and FSR are modeled. Extinction, flame length, HRR, and smoke/aerosol prediction remain future modeling tasks. This boundary preserves a clear distinction between the database's definition of flammability and the narrower scope of the validated machine-learning evidence. The project predicts recorded ignition and FSR outcomes; it does not certify flight materials or solve the full reacting-flow problem. A reliable material-selection tool must make clear both what has been learned from existing tests and where a new experiment remains the only defensible next step.

3 Methodology

3.1 Assembly of the microgravity flammability database

3.1.1 Literature screening and inclusion

The database was assembled before either machine-learning pipeline was designed. Approximately 130 experimental papers and technical reports published from 1966 onward were screened for solid-fuel combustion under microgravity, partial-gravity, or spacecraft-relevant atmospheric and flow conditions. Only about 87 publications exposed sufficiently resolved observations to map ignition or flame-spread rate (FSR) into individual database rows. A paper was useful for modeling only when the experiment, specimen, environment, and measured outcome could be connected at row level; a qualitative conclusion without recoverable conditions could support the literature review but not supervised learning.

The resulting corpus is deliberately broader than one material family or rig. It includes wires, flat specimens, cylinders, and spheres tested on the ground, in drop towers, parabolic aircraft, and spacecraft. Each row retains a full human-readable citation, authors, and DOI or report identifier where available. That provenance is essential because observations from one publication are clustered and because ambiguous values must be checked against the source rather than treated as anonymous numbers.

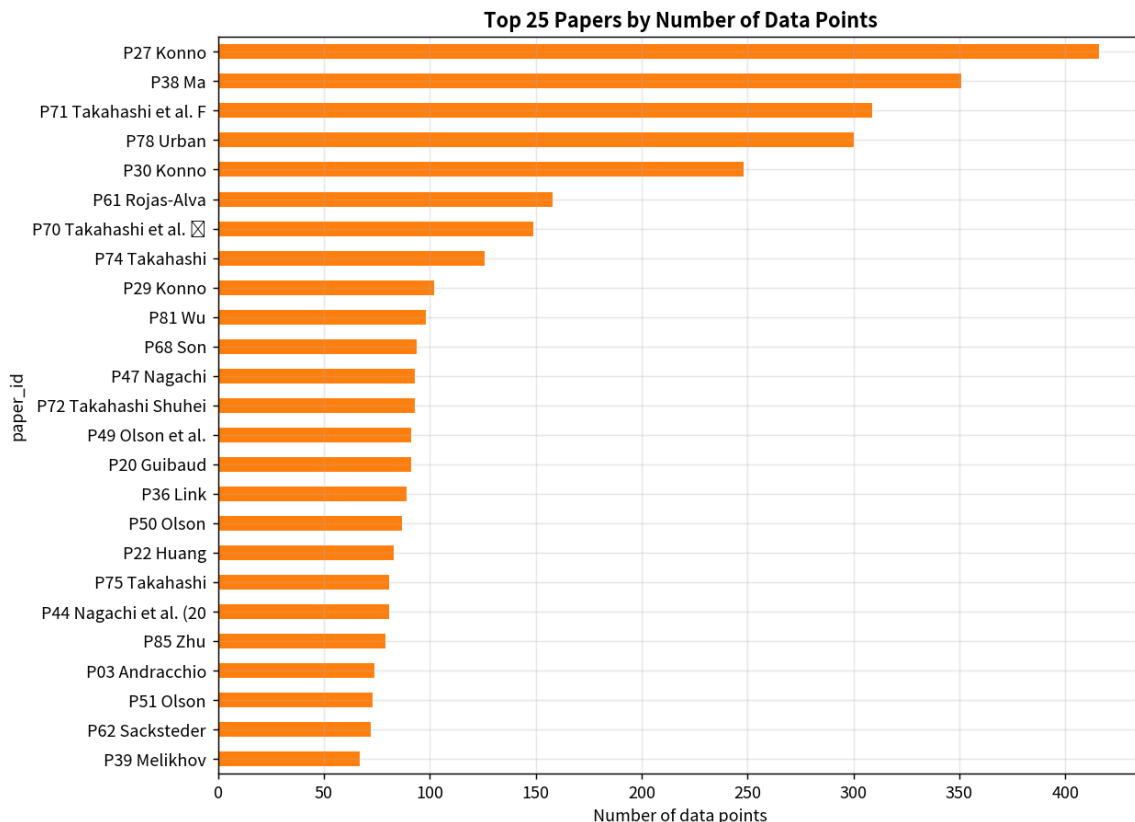


Figure 3.1: The 25 largest publication contributions in the assembled database. The strong variation in rows per paper is a property of the literature, not a balanced experimental design.

The bar chart is steeply ranked: the largest source (Konno) contributes on the order of 400 rows, the next few papers remain above 200, and the 25th paper is already near 70. That shape is why a random row split is optimistic. Conditions from P27 or P38 can appear on both sides of an interpolation split, whereas holding out any one of those papers removes a large, internally correlated block.

3.1.2 Extraction strategy

The first attempted extraction workflow was a conventional optical character recognition (OCR) pipeline. It was not viable for this corpus. The sources range from old scanned papers to long NASA memoranda; notation, units, table structures, and page layouts differ; scans and plotted markers can be blurred; and many results appear only in graphs rather than machine-readable text. Generic OCR also confuses subscripts and symbols, separates values

from their headers, and cannot reliably infer which experimental condition belongs to a plotted point. Applying it at scale would have created a large table with an unacceptably hidden transcription error rate.

The adopted workflow was therefore AI-assisted but human-verified:

1. the source was read to identify the experiment, sample, controlled conditions, reported outputs, and units;
2. an AI extraction pass was used to recover as much structured information as possible for each experimental row;
3. every extracted row was then checked against the paper by the database curator, and missing or incorrectly associated values were completed or corrected manually;
4. units and categorical names were standardized while the original citation and explanatory notes were retained for traceability.

AI was used as an extraction accelerator, not as an authority. This distinction is important for scientific data: a fluent transcription can still attach a correct number to the wrong series, silently flatten a logarithmic axis, or invent precision that the graph does not contain. Manual review remained the acceptance step.

For dense plots, the points were digitized with WebPlotDigitizer [8]. Axis calibration was set from the published ticks, including explicit log-axis selection where required; each color or marker series was traced separately; overlapping points were inspected at high zoom; and the exported coordinates were checked against axis limits and values quoted in the text. This procedure was preferred to AI-internal OCR because exact point coordinates, logarithmic scaling, and overlapping markers were otherwise unreliable.

3.1.3 Row structure, outputs, and quality control

The final schema groups fields into citation, specimen, flow, apparatus, outputs, and notes. Inputs cover geometry, dimensions, fuel and core thermophysical properties, oxygen concentration, diluent, gas properties, pressure, flow, dimensionless transport groups, gravity, facility, and ignition conditions. Flammability is documented through the five outputs introduced in table 1.1: ignition, flame length, FSR, heat-release rate, and smoke/aerosols. Not every paper reports every output.

The raw table uses a category banner as its first row and field names as its second. Values were standardized to SI units wherever the source supported a defensible conversion: length in metres, pressure in pascals, velocity in m/s, power in watts, time in seconds, density in kg/m³, conductivity in W/(m K), heat capacity in J/(kg K), and FSR in m/s. Values whose physical dimension did not match a target field were not coerced merely to fill a cell.

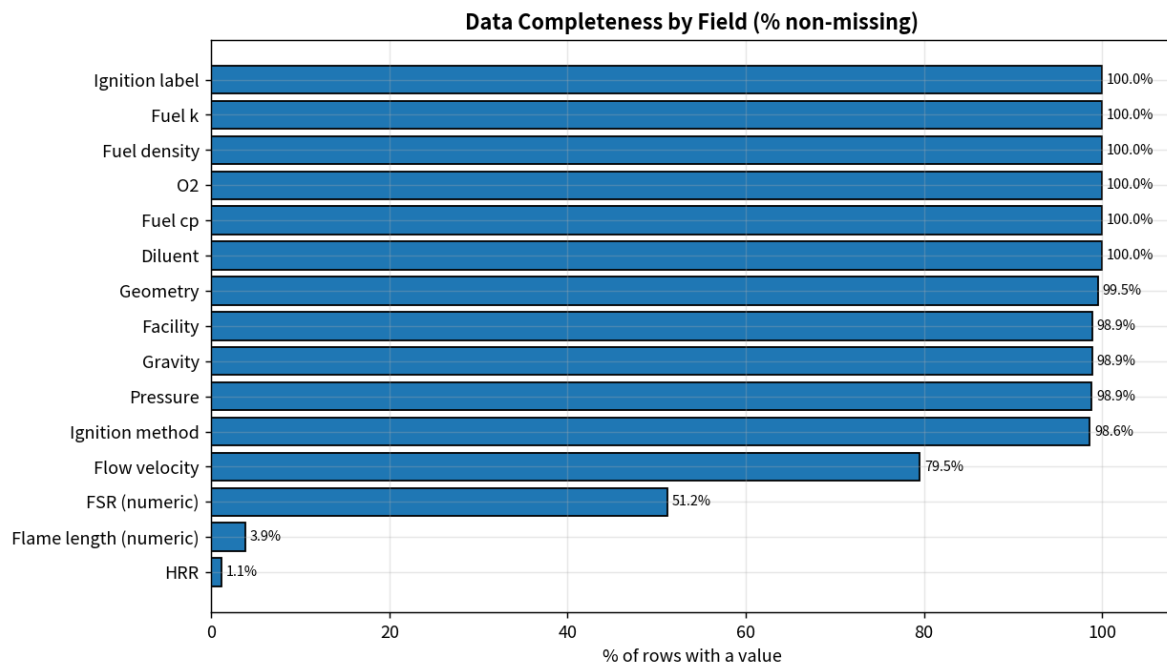


Figure 3.2: Field completeness in the assembled database. Missingness is highly structured: frequently reported atmospheric and material fields coexist with sparse apparatus and output fields.

Ignition labels, fuel thermophysical properties, oxygen, and diluent are essentially complete. Geometry, facility, gravity, pressure, and ignition method remain above 98%. Flow velocity drops to about 80%. Numeric FSR is present in only half of the rows, which is why the regression cohort is smaller than the ignition cohort. Flame length (3.9%) and heat-release rate (1.1%) are too sparse to support a third supervised task. The completeness plot therefore justifies both the two-task scope and the need for fold-local imputation: missingness is a property of what laboratories reported, not a random mask.

Quality control continued in code. Encodings were detected, empty rows and columns removed, target labels normalized, exact modeling duplicates removed, and unparseable values recorded as missing rather than guessed. Canonical paper identity preferred a normalized DOI and otherwise used a normalized citation. This produces identifiers of the form `doi::<normalized-doi>` or `citation::<normalized-citation>`. Cosmetic citation differences therefore cannot place one publication in two paper groups.

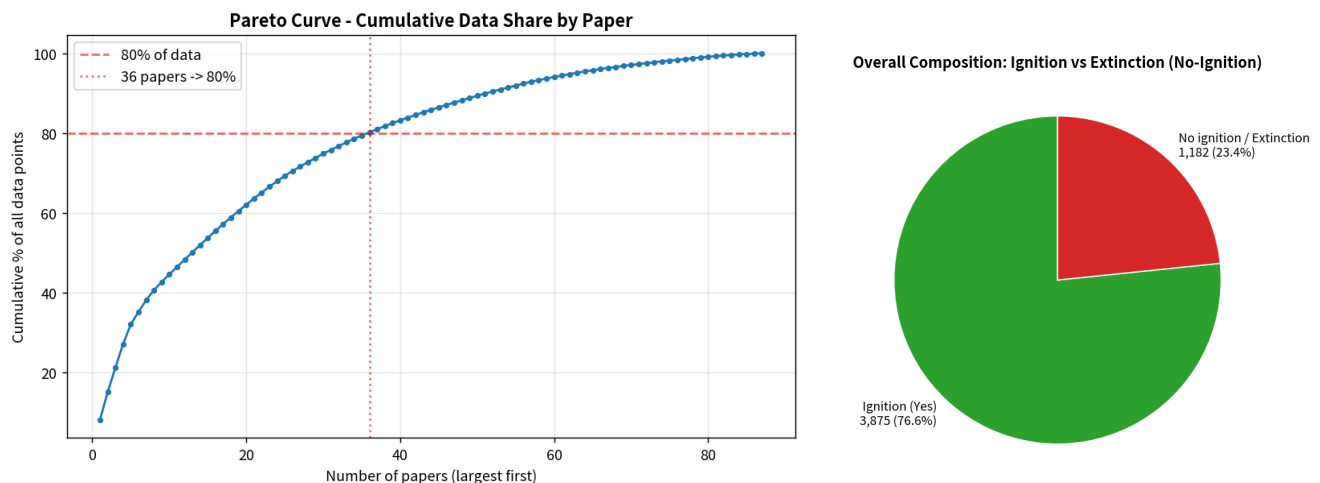
The two tasks use different valid subsets. After feature/target standardization and deduplication, ignition classification contains 4,487 rows from 87 canonical papers (3,383 ignition and 1,104 no ignition). FSR regression contains 2,531 rows with a numeric FSR from 66 papers. The FSR papers contribute between 1 and 193 rows, with median 22.5. Dataset fingerprints are persisted with the splits and evaluations so that a result cannot silently refer to a changed database.

3.1.4 Source-size and class imbalance

The assembled literature table is not a balanced factorial design. Two imbalances dominate every subsequent modeling choice.

The first is source-size imbalance. Publications contribute very different numbers of nearby conditions from a single rig, material batch, or campaign. Figure 3.1 already shows that a handful of papers dominate the row count; the Pareto curve in figure 3.3(a) makes the cumulative effect explicit. After sorting papers from largest to smallest contribution, 36 of 87 papers account for more than 80% of all rows. The remaining long tail contributes the last 20%. A loss that treats every row as an independent draw therefore over-represents densely sampled campaigns and under-represents sparse but physically distinct sources. Paper weighting, paper-grouped evaluation, and paper-cluster bootstrap intervals are the methodological response to this structure.

The second is class imbalance on the ignition label. In the raw assembled table of 5,057 rows, 3,875 observations (76.6%) are ignition and 1,182 (23.4%) are no ignition (figure 3.3(b)). After task-specific cleaning the modeling cohort remains majority-positive: 3,383 ignition versus 1,104 no ignition. A classifier that always predicts ignition would therefore already achieve about three-quarters accuracy and a precision–recall baseline equal to the positive prevalence. Accuracy is not a defensible selection metric. The ignition pipeline instead compares models with ROC–AUC, PR–AUC, and Brier score, and it reports thresholded sensitivity, specificity, precision, F1, balanced accuracy, and Matthews correlation. Class weights and paper weights are tested as candidate recipes rather than assumed to correct the imbalance automatically.



(a) Cumulative data share after sorting papers by contribution. Thirty-six papers supply more than 80% of the rows.

(b) Raw ignition outcome composition: 76.6% ignition versus 23.4% no ignition.

Figure 3.3: Two sampling imbalances in the assembled database. Panel (a) shows that a minority of publications dominate the row count; panel (b) shows that ignition is the majority class. Both motivate grouped evaluation and prevalence-aware classification metrics.

3.2 Methods common to both machine-learning pipelines

3.2.1 Prediction tasks and features

Supervised learning estimates a mapping from measured inputs $\mathbf{x}$ to an observed target y . Here the classifier estimates $\hat{p} = P(y_{\text{ign}} = 1 \mid \mathbf{x})$; the regressor estimates $\hat{v}_{\text{FSR}}$. Paper identifiers, authors, DOI, facility names, free text, and post-outcome quantities were not provided as predictors. Flame length, HRR, and smoke/aerosols were explicitly excluded from both modeling feature sets. The appendix provides the complete feature dictionaries.

The ignition pipeline uses a fixed manifest of 32 numeric and five categorical features. The FSR pipeline detects 29 numeric and four categorical fields from the standardized schema. Both represent fuel and core properties, geometry, oxygen and diluent, pressure, gas transport, flow, gravity, and ignition conditions where available. The persisted FSR manifest also contains categorical `Ignition`; because this conflicts with explanatory documentation, it is disclosed as a schema limitation rather than hidden.

3.2.2 Training, validation, and untouched testing

For a reader new to machine learning, three data roles must remain distinct:

Training data fit model parameters, such as tree splits or neural weights.

Validation data choose hyperparameters, such as tree depth, regularization, or neighborhood size.

Test data measure the selected procedure once on observations not used for either fitting or selection.

The project implements this separation with *nested* evaluation. Each outer split creates a training partition and an untouched test partition. Inside the outer training partition, inner folds compare hyperparameters. Preprocessing, imputation, scaling, one-hot category vocabularies, class or paper weights, and classification thresholds are all learned inside training data. The selected inner configuration is then refitted on the complete outer training partition and evaluated once on the outer test partition. Repeating this process produces a distribution of honest outer-fold scores.

Numeric missing values are replaced by the training-fold median for models that cannot handle missing values; XGBoost retains native numeric missingness. Distance-, margin-, and neural-network models use training-fold standardization. Categorical missingness becomes an explicit “Unknown” level, and one-hot encoding safely ignores a category first encountered in a test fold.

3.2.3 Interpolation versus extrapolation

Both pipelines answer two different questions:

- **Interpolation** holds out rows while allowing other rows from the same paper to remain in training. It asks whether the model predicts new conditions inside campaigns it partly knows.
- **Extrapolation** holds out complete canonical papers. It asks whether the procedure transfers to a new publication, rig, or campaign and is the primary generalization test.

The distinction prevents paper-specific sampling and apparatus signatures from being mistaken for broad physical transfer. All candidates within a task use the same frozen outer assignments. The split files assert full row coverage, no row overlap, and zero paper overlap in grouped protocols.

3.2.4 Candidate comparison, uncertainty, and refitting

The configured portfolios cover XGBoost, random forest or decision tree, k -nearest neighbors (KNN), multilayer perceptron (MLP), and support-vector models. Recipes test different loss functions, paper weights, class weights, target transformations, and bootstrap conditions. The persisted comparison pools are narrower than the YAML portfolios: the merged FSR evidence contains 25 candidates, and the formal ignition selection record omits available MLP shards. Champion claims therefore apply only to candidates in the persisted selection evidence.

Outer-fold means and standard deviations summarize repeat-to-repeat variation. Bootstrap intervals use rows for interpolation and canonical papers for extrapolation, preserving the relevant clustered unit. Only after model selection is complete is the chosen configuration refitted on all labeled rows to create an inference artifact. That full-data model is deployable, but its training fit is not used as evidence of generalization.

3.3 Ignition-classification pipeline

Classification returns a probability rather than a physical yes/no switch. To produce a binary decision, a threshold τ is applied:

$$\hat{y}(\tau) = \begin{cases} 1, & \hat{p} \geq \tau, \\ 0, & \hat{p} < \tau. \end{cases}$$

A lower threshold detects more ignitions but usually increases false alarms; a higher threshold does the reverse. Four operating points are selected from inner out-of-fold predictions: maximum Matthews correlation coefficient (MCC), maximum F1, maximum balanced accuracy, and maximum Youden $J = \text{sensitivity} + \text{specificity} - 1$. The outer test labels are never used to choose τ . Threshold-independent comparison uses ROC-AUC, PR-AUC, and Brier score; thresholded reporting adds sensitivity, specificity, precision, F1, balanced accuracy, and MCC.

3.3.1 Interpolation

Ignition interpolation uses five-fold stratified cross-validation repeated for seeds 42–44, giving 15 outer folds. Stratification keeps the ignition/no- ignition proportion similar in each fold while rows are assigned independently.

Within each outer training partition, stratified inner folds search candidate hyperparameters. Candidates are ordered by mean inner ROC–AUC, then PR–AUC, then lower Brier score. The four thresholds are frozen from the winning inner out-of-fold probabilities before the model is applied to the outer test fold. A separate stratified 80/20 holdout for each seed serves as a quick consistency check, not the primary selection protocol.

Interpolation selection requires valid probabilities, complete folds, passed integrity checks, and ROC–AUC fold standard deviation no greater than 0.15. Candidates within 0.01 of the best ROC–AUC are resolved by PR–AUC, uncertainty, and a declared simplicity order. The resulting champion is then refitted on all ignition-labeled rows for within-domain inference.

3.3.2 Extrapolation

Ignition extrapolation uses `StratifiedGroupKFold`: all rows from one paper stay together while each fold retains both classes where feasible. Five paper-grouped folds repeated across three seeds again produce 15 outer tests. The inner search is also paper-disjoint, preventing a paper held out for inner validation from appearing in the corresponding inner training fold.

The primary metric remains ROC–AUC, but eligibility additionally requires a successful leave-one-paper-out (LOPO) robustness analysis. In LOPO, each of the 87 papers is held out once. A single-class paper cannot define its own ROC–AUC, so LOPO is interpreted through pooled predictions, calibration, and the subset of papers for which a metric is defined. The selected extrapolation model and its thresholds are refitted on the full labeled cohort only after this paper-disjoint comparison is frozen.

3.4 Flame-spread-rate regression pipeline

Regression predicts a continuous FSR in m/s. There is no classification threshold: prediction quality is measured directly in target units. RMSE penalizes larger errors more strongly, MAE reports typical absolute error, R^2 compares residual variance with a mean predictor, and mean bias error records systematic over- or underprediction. MAPE is retained only as a secondary diagnostic because exact and near-zero FSR values make percentage errors unstable.

Every persisted FSR candidate is evaluated on real training data (`rd`) and on `rd_bt`, which appends 1,000 training rows sampled with replacement. Copies inherit their source paper and never enter the test partition. Bootstrap augmentation therefore changes empirical weighting; it does not create a new experiment or extend physical support.

3.4.1 Interpolation

FSR interpolation uses ten repeated, approximately 80/20 row holdouts with seeds 42–51. Target values are binned into six quantiles where feasible so that each split represents the strongly skewed FSR distribution. Papers may occur on both sides. Inside each outer training partition, shuffled K -fold validation compares hyperparameters by lower RMSE, then lower MAE, then higher R^2 . The selected recipe is fitted on all 2,024 training rows and tested on the 507 untouched rows for that repeat.

The selection policy rejects incomplete or non-finite evaluations, RMSE coefficient of variation above 0.45, and mean $R^2 < -0.25$. Candidates within 2% of the lowest RMSE form a tie band resolved by higher R^2 , lower uncertainty, preference for real data, simpler family, and fewer features where the feature sets differ.

3.4.2 Extrapolation

FSR extrapolation also uses ten repeated 80/20 holdouts, but the unit assigned to train or test is the paper. For each repeat, 50 candidate `GroupShuffleSplit` partitions are generated and the most target-balanced paper-disjoint split is retained. Each test contains 14 papers and each training partition 52; row counts vary with paper size. Balance screening reduces avoidable target-range changes without relaxing paper separation.

Inner tuning uses `GroupKFold` by paper. The extrapolation tie-breaks add the interpolation–extrapolation gap so that a recipe is not favored solely for excellent within-paper fit. Cluster bootstrap intervals resample papers rather than thousands of correlated rows. Finally, the winning extrapolation configuration is refitted on all 2,531 FSR rows and stored separately from the outer-fold evidence.

3.5 Explainability and physical interpretation

Explainability is performed after evaluation; it does not influence model selection. Tree impurity importance summarizes how often and how effectively a feature splits training data. Permutation importance measures the loss of held-out performance after one original feature is shuffled. SHAP decomposes each prediction into additive

feature contributions relative to a background:

$$f(\mathbf{x}_i) = \phi_0 + \sum_j \phi_{ij}, \quad I_j^{\text{SHAP}} = \frac{1}{n} \sum_i |\phi_{ij}|.$$

Beeswarm plots retain both sign and feature value, whereas mean absolute SHAP bars show magnitude only.

These are explanations of fitted functions, not causal experiments. Correlated raw and derived features can substitute for one another; missingness can encode paper identity; and a physically plausible ranking does not prove a governing mechanism. The report therefore compares SHAP with held-out permutation importance and interprets repeated patterns as hypotheses for testing and data collection, not as certification evidence.

4 Results

After the preprocessing described in chapter 3, the ignition cohort contains 4,487 labeled rows from 87 papers and the FSR cohort contains 2,531 numeric observations from 66 papers. The figures below report those modeling tasks. Database sampling geometry that is not itself a predictive score—in particular oxygen–pressure coverage—is interpreted in section 5.3 as a map of the experimental search space that remains to be filled.

4.1 Ignition classification

4.1.1 ROC, precision–recall, and the fire-safe operating question

A binary ignition model does not output a physical yes/no switch. It outputs a score $\hat{p} \in [0, 1]$, interpreted as an estimated probability of ignition. Converting that score into a decision requires a threshold τ :

$$\hat{y}(\tau) = \begin{cases} 1, & \hat{p} \geq \tau, \\ 0, & \hat{p} < \tau. \end{cases} \quad (4.1)$$

Each τ produces a confusion-matrix count of true positives (TP), false positives (FP), true negatives (TN), and false negatives (FN). The elementary rates used throughout this section are

$$\text{TPR} = \text{sensitivity} = \text{recall} = \frac{\text{TP}}{\text{TP} + \text{FN}}, \quad (4.2)$$

$$\text{FPR} = 1 - \text{specificity} = \frac{\text{FP}}{\text{FP} + \text{TN}}, \quad (4.3)$$

$$\text{precision} = \frac{\text{TP}}{\text{TP} + \text{FP}}. \quad (4.4)$$

Recall is the fraction of true ignitions that are caught. Precision is the fraction of predicted ignitions that actually ignite. Specificity is the fraction of true non-ignitions that are left unlabeled. No single rate is a complete account of a fire-safety call: missing an ignition (false negative) is the safety-critical error, while flooding a test queue with false alarms (false positives) makes the screener unusable.

Two threshold-independent curves summarize the whole family of operating points. The receiver-operating-characteristic (ROC) curve plots TPR against FPR as τ varies. A no-skill classifier that ranks at random traces the diagonal, with area 1/2. The area under the ROC curve (ROC–AUC) equals the probability that a randomly chosen ignition receives a higher score than a randomly chosen non-ignition. It is a ranking metric: it does not select τ , and it weights the two classes symmetrically through TPR and FPR.

The precision–recall (PR) curve plots precision against recall. Its no-skill baseline is not 1/2; it is the positive prevalence $\pi = P(y = 1)$. In this database $\pi \approx 0.74$ –0.75, so an always-ignite rule already attains precision equal to that prevalence and recall equal to one. The area under the PR curve (PR–AUC, equivalently average precision in figure 4.1) therefore measures how much ranking quality exists *above* that naive majority policy.

A convenient single-threshold summary of the PR trade-off is the F1 score, the harmonic mean of precision and recall,

$$\text{F1} = \frac{2 \text{precision} \cdot \text{recall}}{\text{precision} + \text{recall}}. \quad (4.5)$$

Because a harmonic mean is dominated by the smaller of its two arguments, F1 is large only when both precision and recall are large. It does not, however, use true negatives, so a majority-positive task can reward a low τ that labels almost everything as ignition. Complementary summaries that do use the full confusion matrix are balanced accuracy (sensitivity + specificity)/2, Youden’s $J = \text{sensitivity} + \text{specificity} - 1$, and the Matthews correlation

$$\text{MCC} = \frac{\text{TP TN} - \text{FP FN}}{\sqrt{(\text{TP} + \text{FP})(\text{TP} + \text{FN})(\text{TN} + \text{FP})(\text{TN} + \text{FN})}}. \quad (4.6)$$

Thresholds that maximize these criteria are selected inside training folds and are interpreted as operational limitations in section 5.1; they are not used to claim a unique “best” fire-safety setting.

Figure 4.1 shows both curves for the selected XGBoost classifier on a held-out partition. The ROC curve rises steeply toward the upper-left corner and attains AUC 0.897, well above the chance diagonal. That result says the model ranks ignitions above non-ignitions for most pairs. It does *not* by itself say that a fire-safe call is available: a high ROC–AUC can coexist with many false positives when negatives are the minority class, because FPR is computed only among true non-ignitions.

The PR curve is the more informative fire-safety diagnostic for exactly that reason. Precision remains near one up to recall of about 0.7, then declines as the threshold is lowered to recover the remaining ignitions. Average

precision is 0.963, far above the prevalence baseline of 0.74 drawn as the dashed line. In operational language: there exists a high-precision region in which flagged cases are very likely to be true ignitions, and the model can recover a large majority of ignitions before precision collapses. Pushing recall toward one—the instinctive “never miss a fire” policy—still costs precision, which is why PR-AUC, rather than ROC-AUC or accuracy, is the curve that should govern a screening claim on this imbalanced label. ROC-AUC remains useful as a threshold-free ranking score for model comparison; PR-AUC is the quantity that asks whether positive-class decisions remain reliable.

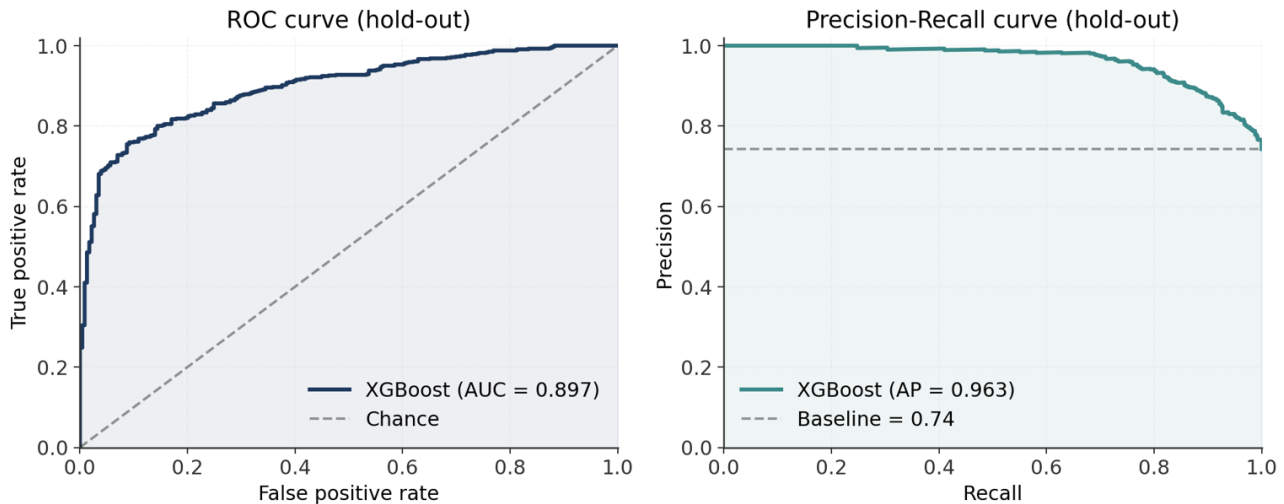


Figure 4.1: Held-out ROC and precision-recall curves for the ignition XGBoost classifier. The ROC panel (left) plots true-positive rate against false-positive rate; the shaded area is ROC-AUC = 0.897, against a chance diagonal of 0.5. The PR panel (right) plots precision against recall; the shaded area is average precision = 0.963, against a no-skill baseline equal to the positive prevalence (0.74). The PR curve is the more relevant fire-safety summary because ignition is the majority class and because a screening call is a statement about the positive class.

4.1.2 Interpolation and extrapolation by model family

The selected interpolation classifier is the paper-weighted XGBoost recipe `xgb_paper_sqrt`. Across 15 repeated stratified outer folds it reaches ROC-AUC 0.9076 ± 0.0102 , PR-AUC 0.9687 ± 0.0039 , and mean Brier score 0.1063. The selected extrapolation classifier is the unweighted `xgb_baseline`. Its paper-grouped ROC-AUC is 0.7183 ± 0.0444 , PR-AUC 0.8767 ± 0.0298 , and Brier score 0.1791. The paper-cluster bootstrap estimate of grouped ROC-AUC is 0.7101 with 95% interval 0.6408–0.7848; leave-one-paper-out robustness is similar (0.7020, 0.6376–0.7751).

Figure 4.2 presents the same two questions for the best persisted candidate in each family, so interpolation and extrapolation are read together rather than as separate leaderboards. Panel (a) uses ROC-AUC; panel (b) uses PR-AUC. In every family the interpolation marker lies to the right of the extrapolation marker, with much smaller error bars. XGBoost is the strongest family on ROC-AUC for both protocols (interpolation 0.901 ± 0.002 , extrapolation 0.718 ± 0.044 on the plotted representatives). SVM and MLP occupy a middle band; decision trees and k -nearest neighbors are weaker, and the decision-tree extrapolation bar is the most unstable ($SD = 0.100$).

PR-AUC preserves the same family ranking for interpolation, with XGBoost at 0.966 ± 0.001 . Under extrapolation the picture is slightly different: SVM reaches 0.879 ± 0.022 and XGBoost 0.877 ± 0.030 , so the two families are indistinguishable on the positive-class ranking metric once papers are held out. That is consistent with the fire-safety reading of figure 4.1: ROC-AUC and PR-AUC can disagree about a close call because they answer different questions. All families remain well above the prevalence baseline of ≈ 0.75 , but the drop from interpolation (PR-AUC ≈ 0.93 – 0.97) to extrapolation (≈ 0.82 – 0.88) is the quantity that limits transfer to a new campaign. Tree-depth flexibility (XGBoost versus a single decision tree) and margin-based models (SVM) retain more of that positive-class ranking than memory-based KNN or an unpruned tree.

Probability ranking is not calibration. Figure 4.3 compares mean predicted probability with the observed ignition fraction. The interpolation champion tracks the identity line closely, especially above probability 0.5. The extrapolation champion is miscalibrated: its scores occupy a narrower, higher band, and high predicted probabilities overstate the observed ignition rate. A numerical probability from the grouped model therefore cannot be read as a frequentist ignition risk without a source-conditional calibration step.

4.1.3 Ignition explainability

The held-paper XGBoost SHAP plot in figure 4.4 shows both contribution direction and feature value. Oxygen has the largest global SHAP magnitude, followed by absolute flow speed, Prandtl number, gas molar mass, fuel

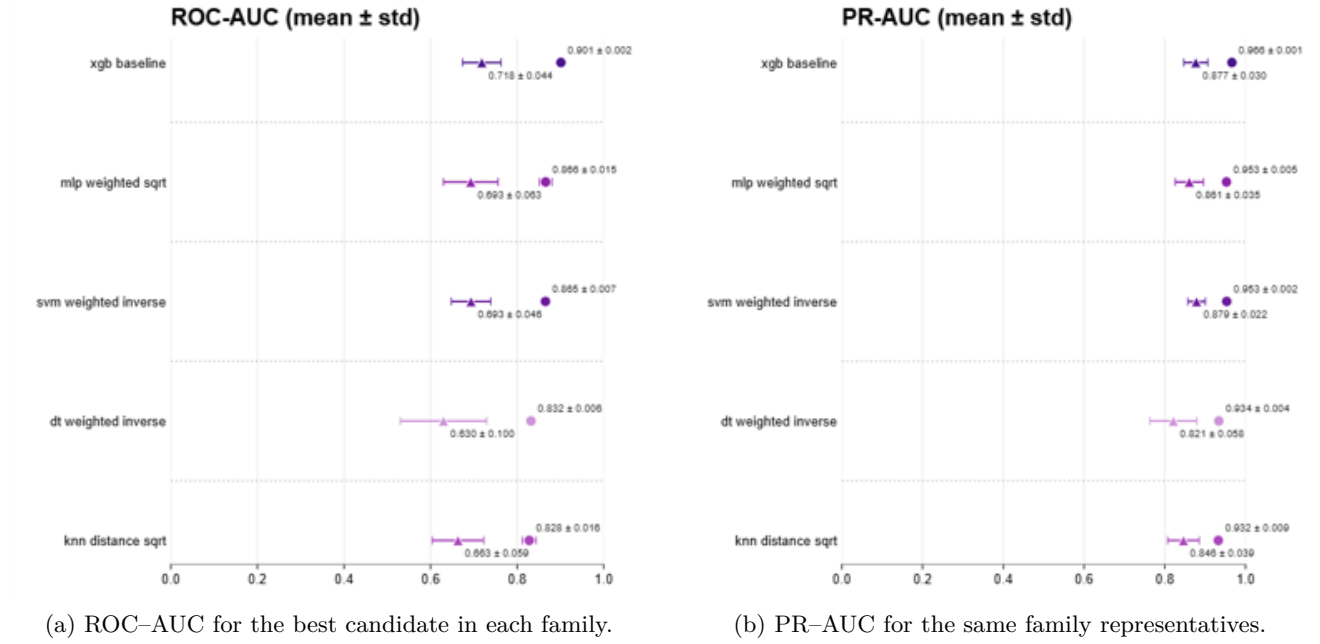


Figure 4.2: Ignition interpolation (circles) versus paper-grouped extrapolation (triangles), mean $\pm$ standard deviation. Every family loses performance when entire papers are withheld, and extrapolation uncertainty is systematically larger. XGBoost leads ROC-AUC in both protocols; under PR-AUC the extrapolated SVM and XGBoost candidates are essentially tied. The two panels together replace a separate “generalization-gap” ranking: the gap is the horizontal separation between the two markers of each family.

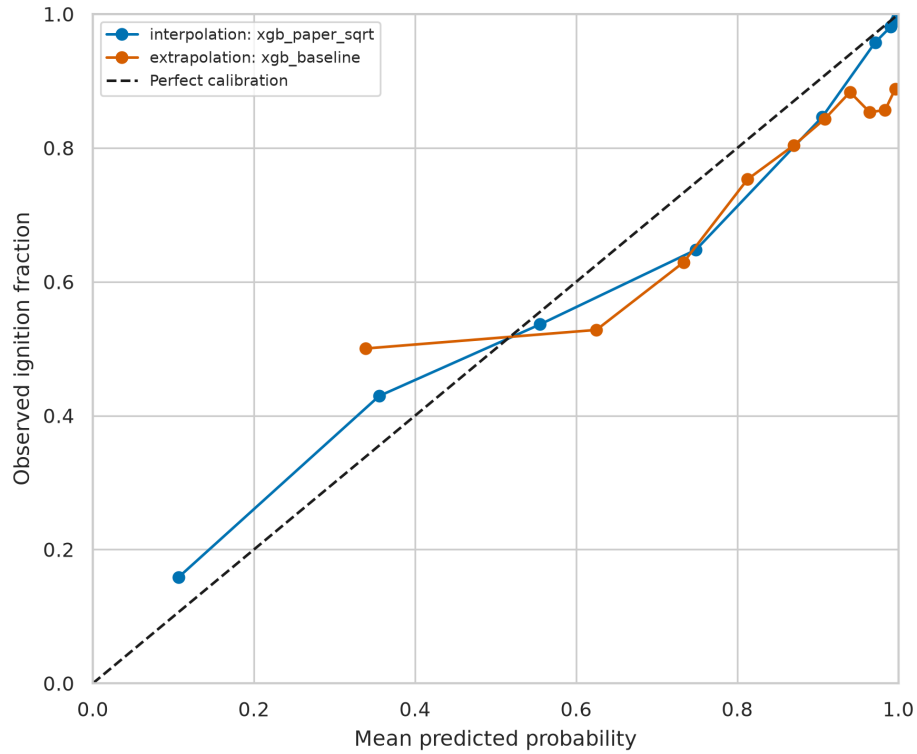


Figure 4.3: Reliability diagram for the interpolation and extrapolation ignition champions. Perfect calibration is the dashed diagonal. The interpolation curve follows the identity; the extrapolation curve is compressed toward high scores and lies below the diagonal at the upper end, indicating over-confident high-risk predictions on held-out papers.

pyrolysis temperature, and ignition time. Higher oxygen values generally move predictions toward ignition in this fitted model. Low absolute flow speed produces large negative SHAP values, consistent with a quenching branch in which weak transport cannot sustain a flame. Long recorded ignition times also push the score downward, while several transport quantities show nonlinear or mixed patterns.

Permutation importance gives a different ranking: fuel pyrolysis temperature, ignition time, fuel heat capacity, and thermal diffusion time produce the largest positive held-out ROC–AUC losses when shuffled. Oxygen has high SHAP magnitude but is not the leading permutation feature in this held-out sample. The disagreement is evidence of correlated and substitutable predictors, not a contradiction and not a causal ranking. Oxygen, flow, and thermal time scales nonetheless recur across both views, which is the level of interpretation the observational table can support.

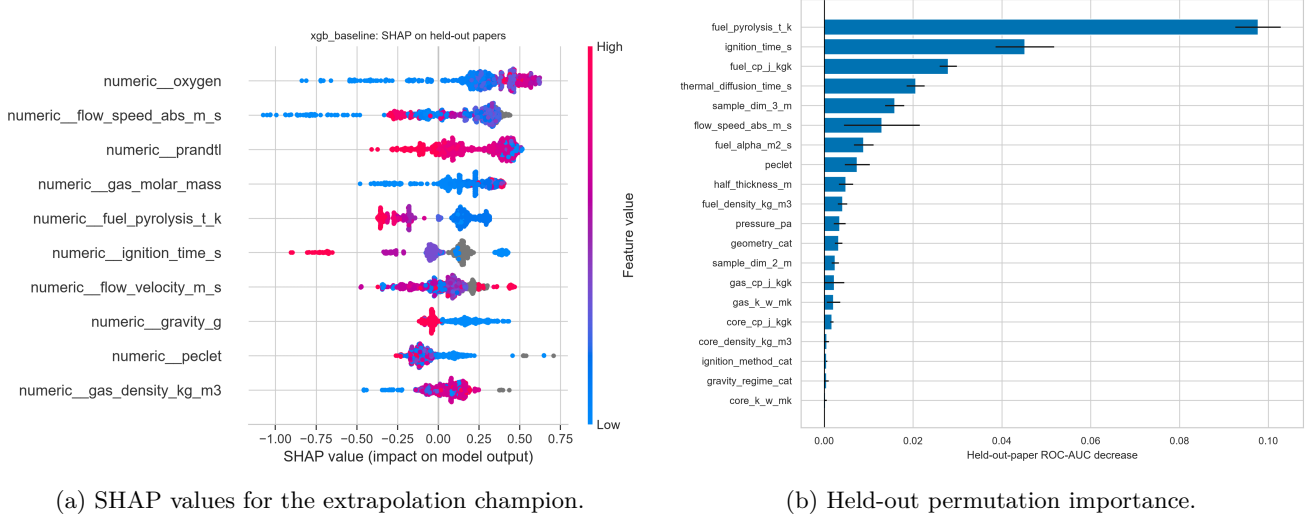


Figure 4.4: Ignition explainability on held-out papers. The beeswarm retains sign and feature value; the permutation bars measure the loss of ROC–AUC after shuffling one original feature. Repeated oxygen, flow, and thermal variables are predictive associations in the fitted function, not identified mechanisms.

4.2 Flame-spread-rate regression

4.2.1 Interpolation and extrapolation together

Regression has no classification threshold: the model predicts a continuous FSR in m/s, and quality is measured in those units. Two summaries are used throughout. RMSE is the root-mean-square error

$$\text{RMSE} = \sqrt{\frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2}, \quad (4.7)$$

which penalizes large misses and remains dimensionally an FSR. The coefficient of determination

$$R^2 = 1 - \frac{\sum_{i=1}^n (y_i - \hat{y}_i)^2}{\sum_{i=1}^n (y_i - \bar{y})^2} \quad (4.8)$$

compares residual variance with a mean predictor; $R^2 = 1$ is a perfect fit and $R^2 = 0$ is no better than predicting $\bar{y}$. The two metrics are not interchangeable. RMSE is the engineering error in physical units; R^2 is a scale-free statement about explained variance and is sensitive to the target spread of the held-out papers. Ranking models by RMSE can therefore disagree with ranking them by R^2 when folds differ in variance, which is why both axes appear below.

The unaugmented random-forest baseline is selected for both protocols, with separately tuned configurations. Across ten repeated row holdouts it reaches $R^2 = 0.8790 \pm 0.0178$, RMSE 0.006263 ± 0.000472 m/s, and MAE 0.002103 ± 0.000173 m/s. Across ten paper-grouped holdouts the same family reaches $R^2 = 0.4576 \pm 0.2215$, RMSE 0.012205 ± 0.004115 m/s, and MAE 0.004819 ± 0.001144 m/s. Grouped RMSE is 1.95 times the interpolation value; the paper-cluster 95% interval for grouped RMSE is 0.006525–0.017839 m/s. Bootstrap-augmented XGBoost has a numerically slightly lower interpolation RMSE (0.006236 m/s) but falls within the 2% tie band and has greater fold variability, so the declared tie policy favors the unaugmented random forest.

Figure 4.5 shows every persisted candidate under both protocols and both training conditions (real data versus bootstrap-augmented training). Interpolation bars (blue) are consistently lower than grouped bars (orange). Tree

ensembles occupy the left-hand, low-RMSE group; KNN occupies a middle band; SVR and MLP occupy the right-hand, high-error group, with grouped error bars that can exceed 0.03 m/s. Augmentation does not close the protocol gap: the dark and light bars of a given protocol remain close relative to the blue–orange separation. The figure therefore answers interpolation and extrapolation in one comparison: family identity and evaluation protocol dominate recipe-level tweaks.

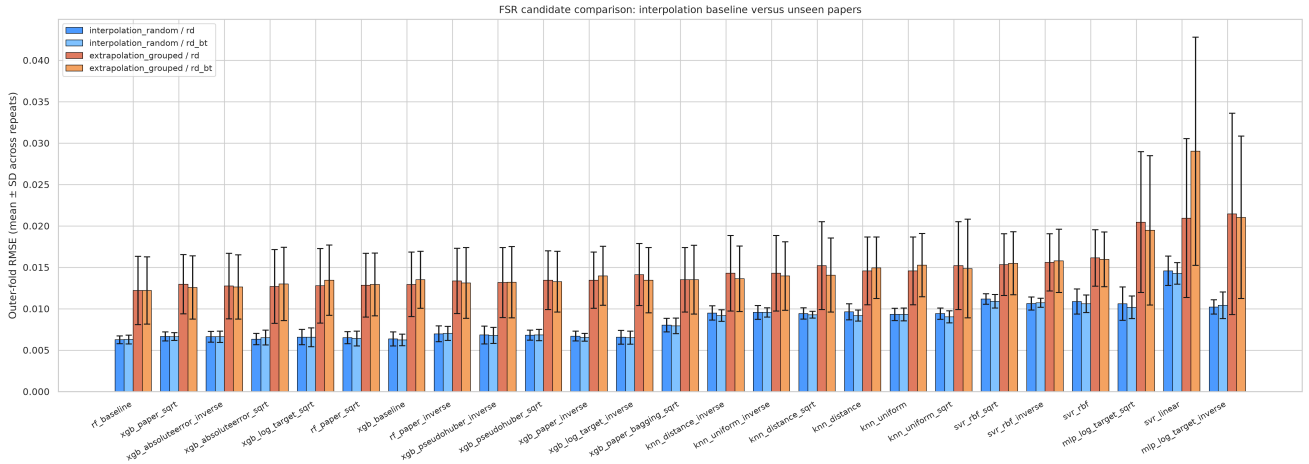


Figure 4.5: FSR outer-fold RMSE for all persisted candidates. Blue bars are row-level interpolation; orange bars are paper-grouped extrapolation; paler and darker shades distinguish real versus bootstrap-augmented training. Tree ensembles form a leading cluster; KNN, SVR, and MLP form successively weaker clusters. The interpolation–extrapolation gap is larger than the gap among leading tree recipes.

Figure 4.6 makes the two ranking metrics explicit. Each point is a candidate; the horizontal axis is mean R^2 (higher is better) and the vertical axis is mean RMSE (lower is better), so the desirable corner is the bottom right. A dashed line separates grouped extrapolation (left, purple) from random interpolation (right, blue). Three facts follow immediately.

First, the two metrics largely agree inside each protocol: points fall along a descending cloud, so improving R^2 generally reduces RMSE. There are nevertheless two legitimate ways to rank models. RMSE is the quantity to quote for a physical error budget; R^2 is the quantity to quote when asking whether the model beat a mean predictor on that fold’s target spread. A candidate can look competitive on one axis and less so on the other if its errors concentrate on high-variance papers.

Second, model families are visually separated in both panels. Random forest and XGBoost occupy the lower-right cluster of each protocol; KNN occupies an intermediate cluster; SVR and MLP occupy the upper-left cluster. Family identity is therefore a stronger organizing principle than individual hyperparameter variants. The highlighted random-forest baseline ($R^2 = 0.879$, RMSE = 0.00626 m/s in interpolation; $R^2 = 0.458$, RMSE = 0.01221 m/s in extrapolation) sits at the front of the tree cluster in both tasks.

Third, the protocol shift is a translation of the entire family structure, not a reordering. The same relative ranking survives paper hold-out, but the absolute location moves from $R^2 \approx 0.4\text{--}0.9$ to $R^2 \approx 0.05\text{--}0.46$ and from RMSE $\approx 0.006\text{--}0.015$ to RMSE $\approx 0.012\text{--}0.016$ m/s. Interpolation competence does not imply transfer; family ranking, however, is stable enough that a screening tool can prefer tree ensembles without pretending that within-paper $R^2 = 0.88$ is available for a new campaign.

4.2.2 Held-out parity and error structure

The two panels of figure 4.7 diagnose where the grouped random forest succeeds and where it fails. Panel (a) is the extrapolation parity plot: predicted versus experimental FSR, with the one-to-one line as the reference. The pooled snapshot on that figure is $R^2 = 0.425$ and RMSE = 0.0129 m/s, consistent with the repeated-holdout mean $R^2 = 0.458 \pm 0.222$. Dense low-FSR observations lie near the diagonal. Above about 0.05 m/s the cloud peels away below the line: experimental values continue to 0.21 m/s while predictions rarely exceed 0.08 m/s. The model regresses toward the densely sampled low-spread mean.

Panel (b) plots the same predictions as error (predicted minus experimental) against true FSR. At near-zero spread the errors are small and slightly positive: the model overpredicts the slowest flames. As experimental FSR increases, the error trend is a descending line that reaches almost -0.20 m/s at the fastest observations, meaning a high-spread event is predicted near zero. Error variance also widens with true FSR (heteroscedasticity). For a fire-safety regressor this pattern is the opposite of a conservative bias: the dangerous tail is underpredicted. The hardest held-out paper reaches RMSE 0.1331 m/s, whereas many papers remain below 0.003 m/s. A single pooled R^2 therefore conceals campaign-level failure modes that the error plot makes visible.

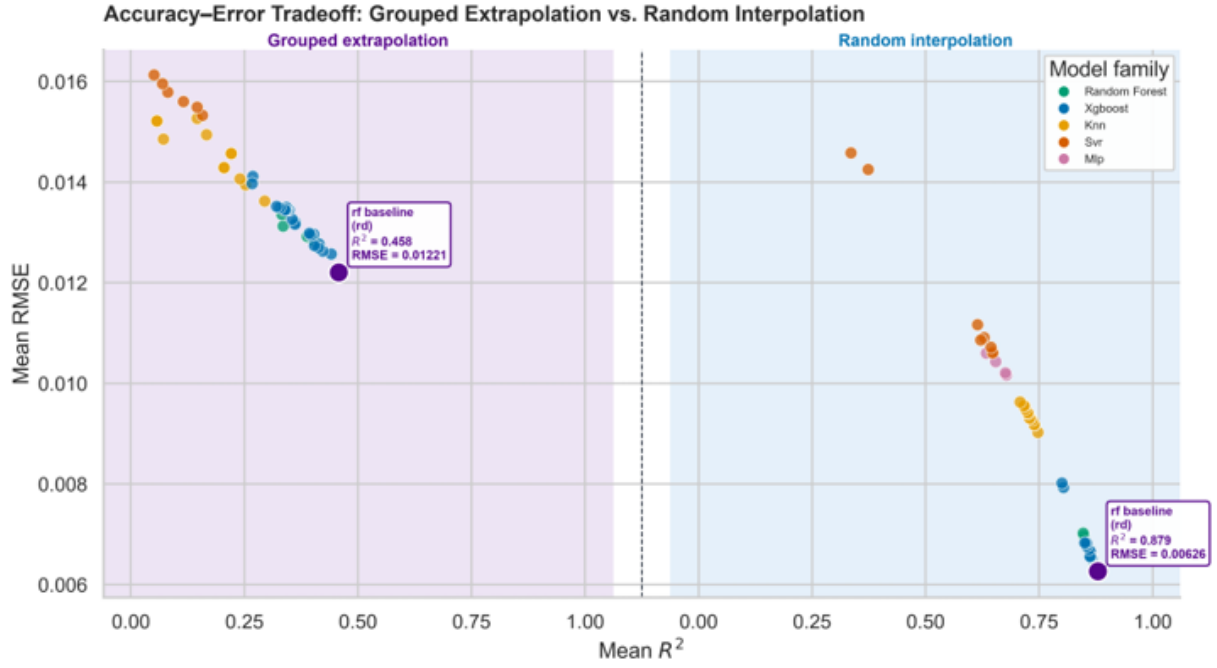
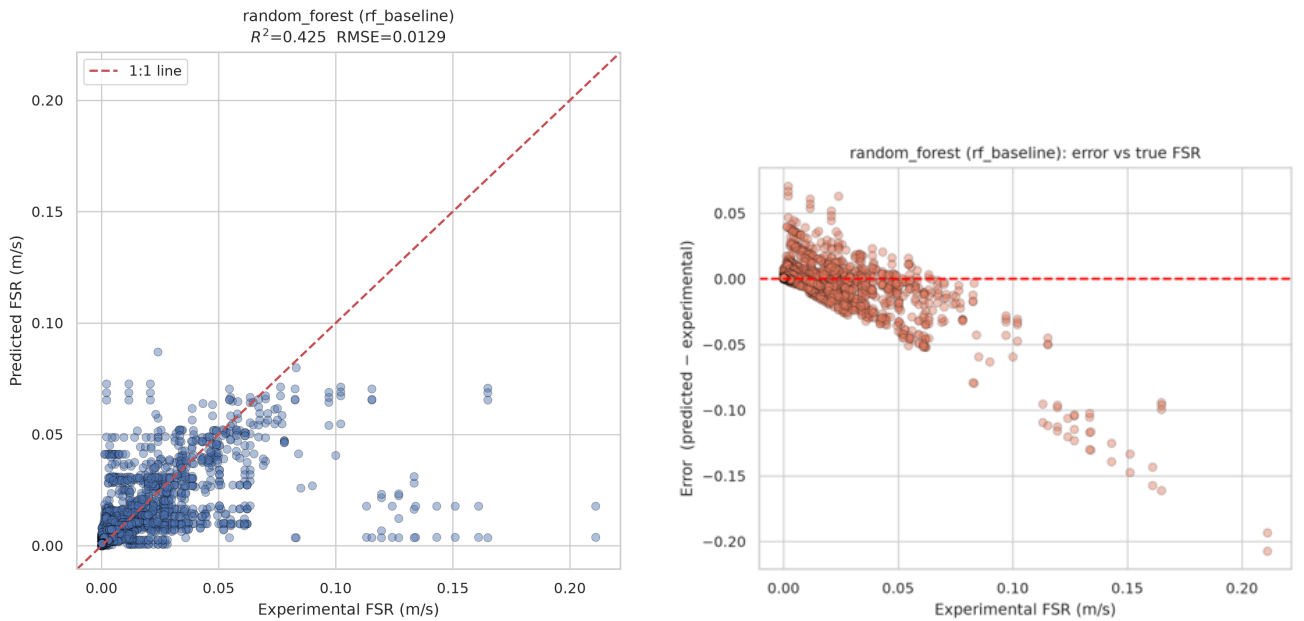


Figure 4.6: Accuracy–error trade-off for FSR candidates. Mean R^2 increases to the right and mean RMSE decreases downward, so better models lie toward the bottom-right of each panel. Colour denotes model family. Grouped extrapolation (left) and random interpolation (right) show the same family separation—random forest and XGBoost, then KNN, then SVR and MLP—shifted by the protocol. The highlighted points are the selected random-forest baseline.



(a) Predicted versus experimental FSR on held-out papers. Points below the dashed one-to-one line are underpredictions. (b) Error (predicted minus experimental) versus true FSR. The descending trend is systematic underprediction of fast spread.

Figure 4.7: Grouped random-forest diagnostics. Panel (a) is the extrapolation R^2 view: low-FSR points hug the identity, while high-FSR points fall below it. Panel (b) is the corresponding residual view. Together they show regression toward the low-spread majority and growing error in the sparse high-FSR tail.

4.2.3 FSR explainability

The FSR random forest relies most strongly on thermal diffusion time, oxygen, gas thermal conductivity, and flow velocity in its SHAP summary (figure 4.8). The beeswarm adds direction: high oxygen generally pushes FSR upward, while large thermal diffusion times—a thick or thermally inertial solid—push FSR downward. Held-out permutation importance independently identifies oxygen, flow velocity, gas conductivity, and thermal diffusion time as the largest positive RMSE degradations.

Several correlated material variables have negative permutation importance in the realized held-out sample. That means shuffling happened to reduce error; it does not imply a physically protective effect. The repeated appearance of oxygen, flow, and thermal transport across two explanation methods supports their use as hypotheses and coverage priorities, not as causal coefficients.

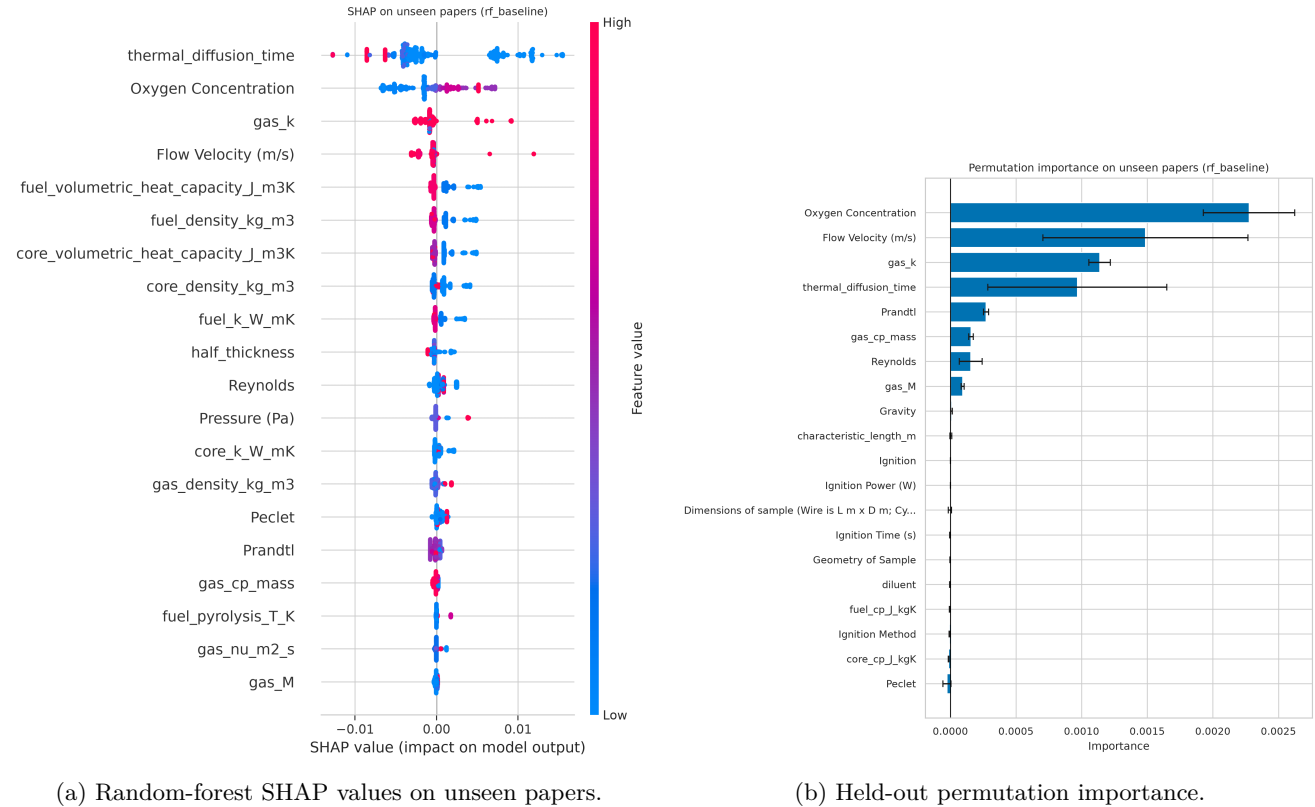


Figure 4.8: FSR explainability on unseen papers. Thermal diffusion time and oxygen dominate SHAP; permutation importance recovers oxygen, flow, gas conductivity, and thermal diffusion time as the features whose shuffling most increases RMSE.

4.3 Synthesis

The figures establish one common result across two targets and different model families: within-paper interpolation is strong, while transfer to an unseen paper is moderate and substantially more uncertain. Ignition ROC-AUC falls from approximately 0.91 to 0.72, and PR-AUC from 0.97 to 0.88; FSR RMSE approximately doubles and R^2 falls from 0.879 to 0.458. Family identity is stable: tree ensembles lead both tasks on both protocols. These protocol effects are larger than the differences among the leading tree recipes.

PR-AUC, not ROC-AUC, is the ignition summary that matches a fire-safe screening call, because the label is majority-positive and the operational question concerns the ignition class. For FSR, RMSE and R^2 are two legitimate ranking axes that largely agree, while the residual plot shows that pooled scores hide underprediction of the high-spread tail. The defensible result is a screening tool with explicit domain and uncertainty limits, not a universal flammability law.

5 Discussion

The central result of this study is not that one algorithm “solves” material fire safety, but that the apparent predictive difficulty depends strongly on what is held out. When rows were divided at random, experiments from the same publication could occur in both training and test sets. Under that interpolation question, the flame-spread-rate (FSR) random-forest champion reached $R^2 = 0.879 \pm 0.018$ with an RMSE of $0.00626 \pm 0.00047 \text{ m s}^{-1}$, while the ignition XGBoost champion reached a ROC-AUC of 0.908 ± 0.010 . When complete papers were held out, the corresponding results fell to $R^2 = 0.458 \pm 0.222$ and RMSE $0.01221 \pm 0.00412 \text{ m s}^{-1}$ for FSR, and ROC-AUC 0.718 ± 0.044 for ignition. The FSR error approximately doubled, and the ignition ROC-AUC decreased by about 0.19. This consistent row-versus-paper generalization collapse is the most consequential scientific finding. It indicates that models can interpolate effectively among conditions resembling a campaign already represented in training, but only partially transfer to an unseen publication, apparatus, laboratory practice, or sampled regime.

This chapter interprets that finding under four deliberately visible allocations: limitations (approximately 30%), comparison with the literature (approximately 30%), real-world applications (approximately 30%), and future work (approximately 10%). These proportions reflect the practical importance of defining what the models do not establish, placing their scores beside protocol-compatible evidence, and translating them into a cautious decision-support role.

5.1 Limitations

5.1.1 Database heterogeneity and the unit of generalization

The source database is a synthesis of published experiments rather than a prospectively designed factorial study. The FSR analysis contains 2,531 usable target rows from 66 canonical papers; the ignition analysis contains 4,487 deduplicated rows from 87 papers, of which 3,383 report ignition and 1,104 report no ignition. These counts provide considerably more observations than a single campaign, but rows are not independent replications of a common data-generating process. A publication may contribute many nearby settings from one rig, material batch, diagnostic method, gravity platform, or laboratory. Another may contribute only a handful of observations in a distinct regime. Consequently, a row-level split can place near-neighbours—and campaign-specific signatures—on both sides of the evaluation boundary.

The grouped protocol addresses the clearest version of this dependence by constructing canonical paper identities from DOI or citation and prohibiting paper overlap. It therefore tests a harder and more relevant question: transfer to a publication that was absent from model fitting and inner hyperparameter selection. It does not, however, guarantee transfer to every future spacecraft material or operating envelope. Papers can share laboratories, investigators, facilities, wire stocks, reporting conventions, or inherited correlations. Conversely, two campaigns in one paper may differ materially. “Paper” is thus a defensible, reproducible proxy for experimental source, not a perfect ontological definition of domain. A still stronger validation would hold out laboratories, facilities, material families, gravity platforms, or time periods, but the current metadata do not support all such partitions consistently.

The collapse under paper holdout is evidence that publication identity carries predictive structure beyond the intended physical variables. That structure may include legitimate regime information omitted from the schema, such as apparatus heat losses or material morphology, as well as source-specific measurement and reporting practices. The result should not be described simply as overfitting in the narrow algorithmic sense. It is more accurately a combination of source shift, incomplete covariate description, clustered sampling, and possible outcome-definition differences. The large FSR fold variability— R^2 ranging from approximately 0.018 to 0.746 for the grouped champion—further shows that transfer difficulty depends on which papers are withheld.

5.1.2 Missingness, measurement comparability, and representation

Literature synthesis inevitably produces sparse and uneven feature coverage. Material thermophysical properties may be tabulated, inferred from material labels, or absent. Geometry can be reported in formats that are not directly commensurate across wires, flat samples, cylinders, and spheres. Pressure, flow, oxygen concentration, ignition settings, and gravity are not varied independently, and some are fixed or omitted within individual campaigns. The preprocessing contract handles this computationally: numeric values are median-imputed for most families, categorical missingness is assigned an “Unknown” level, and XGBoost can use native numeric missingness. Computational handling is not equivalent to recovering the unobserved physics.

Missingness may itself be source-dependent. If one laboratory systematically reports a property and another does not, a tree can use the missingness pattern as a source fingerprint. This can improve random-split prediction while harming transfer. Median imputation also compresses genuinely different unknown cases to a common value, while a constant categorical level conflates “not measured”, “not applicable”, and “not reported”. The grouped results therefore integrate both the difficulty of physical extrapolation and the consequences of imperfect harmonisation.

Feature engineering improves consistency but does not remove these concerns. The ignition pipeline uses 32 numeric and five categorical features, including material, gas-transport, operating-condition, geometry, flow-direction, gravity-regime, and ignition-method descriptors. The FSR pipeline detects the available numeric and categorical roles after excluding explicit source identifiers, duplicate target fields, and post-outcome variables. Derived quantities such as Reynolds, Peclet, Prandtl, volumetric heat capacity, thermal diffusivity, and thermal diffusion time can encode useful physical structure. Yet derived features share inputs and can be mutually dependent; their values are only as reliable as the source measurements and assumptions used to calculate them.

One schema discrepancy requires explicit qualification: the persisted FSR feature manifest includes categorical **Ignition**, although the pipeline documentation describes ignition fields as excluded. Its reported tree and SHAP importance is zero, but that does not substitute for a controlled ablation. Future FSR evaluation should remove the field and regenerate frozen comparisons before asserting that the documented and executed schemas are identical.

5.1.3 Imbalance and observational sampling

Two forms of imbalance matter. First, ignition is a majority-positive task: approximately three quarters of the 4,487 records are ignition cases. A high PR–AUC is therefore partly facilitated by prevalence and must be read beside ROC–AUC, Brier score, sensitivity, specificity, and threshold-dependent performance. Second, papers contribute unequal row counts. A densely sampled campaign can dominate empirical risk even if it represents a narrow region. The candidate portfolios test no, square-root, and inverse paper weights, and weight-blind families use deterministic probability-proportional resampling. These mechanisms change each source’s influence, but cannot manufacture missing regimes. The selected grouped champions were the unweighted baselines: random forest for FSR and XGBoost for ignition. This does not prove that source imbalance is unimportant. It only shows that the tested weighting corrections did not improve the selection objective enough under these frozen splits.

Observational coverage creates a more fundamental limitation. Oxygen, pressure, gravity, flow, geometry, material properties, and ignition practice were chosen by investigators for particular scientific questions. Their joint distribution is therefore neither random nor uniform. For example, a material may occur mainly in one geometry and gravity regime, while a pressure range may be associated with one facility. A flexible learner can predict from these correlations without separating their physical effects. Sparse partial-gravity observations are especially problematic because lunar and Martian gravity conditions cannot be treated as densely sampled interpolation domains merely because gravity is a numeric feature.

5.1.4 Threshold policies and the operating points they induce

A ROC or PR curve ranks cases; a deployed screener still needs a number τ at which a probability becomes a yes. Four policies were frozen from inner out-of-fold scores: maximum MCC, maximum F1, maximum balanced accuracy, and maximum Youden J . Balanced accuracy and Youden J selected the same thresholds in both champions, so figure 5.1 shows three distinct policies for each of the interpolation and extrapolation models.

The figure plots sensitivity (recall) against specificity. The dashed diagonal is the locus of equal sensitivity and specificity; points above it catch ignitions at a higher rate than they correctly clear non-ignitions, and points below it do the reverse. Marker shape distinguishes the interpolation champion (`xgb_paper_sqrt`, circles) from the extrapolation champion (`xgb_baseline`, triangles). Colour distinguishes the policy.

Interpolation occupies the upper-right quadrant. At $\tau \approx 0.84$ the balanced-accuracy policy attains sensitivity 0.77 and specificity 0.90; MCC at $\tau \approx 0.79$ is similar (0.79, 0.87). Those high thresholds are usable screening settings inside known campaigns: most ignitions are still caught, and most true non-ignitions remain unlabeled. The F1 policy drops the threshold to $\tau \approx 0.37$ and induces a different operating point: sensitivity 0.95 but specificity only 0.48. Because F1 ignores true negatives, it buys almost-complete recall by labeling many non-ignitions as ignition. Even within interpolation, F1 is therefore not a fire-safe default; it is a high-recall policy whose false-alarm cost must be accepted explicitly.

Extrapolation induces a much harsher map. All three triangles lie on or below the diagonal. Balanced accuracy at $\tau \approx 0.86$ sits on the diagonal at (0.66, 0.66): the grouped champion, once forced to a single cut, is no better than a balanced coin in this sensitivity–specificity plane. MCC at $\tau \approx 0.81$ trades a little more recall (0.73) for weaker specificity (0.58). F1 at $\tau \approx 0.29$ produces sensitivity 0.98 and specificity 0.06: the model essentially predicts ignition for every held-out case. A “never miss a fire” threshold on a new paper therefore does not yield a selective screener; it yields a near-constant positive alarm.

Two consequences follow. First, no threshold trained on interpolation transfers as a universal operating point: the same policy names produce qualitatively different (sensitivity, specificity) pairs once papers are withheld. Second, metric choice is itself a safety decision. Optimizing F1, or any recall-heavy scalar, induces an alarm-almost-always rule under source shift, which would saturate a test campaign rather than prioritize it. Optimizing balanced accuracy or MCC induces a more even trade-off, but under extrapolation that trade-off is only moderate and is not a certification margin. The curves in figure 4.1 remain valid rankings; figure 5.1 shows that converting those rankings into a binary fire-safe call still requires an explicit loss function, and that the grouped evidence does not currently support a high-recall, high-specificity cut.

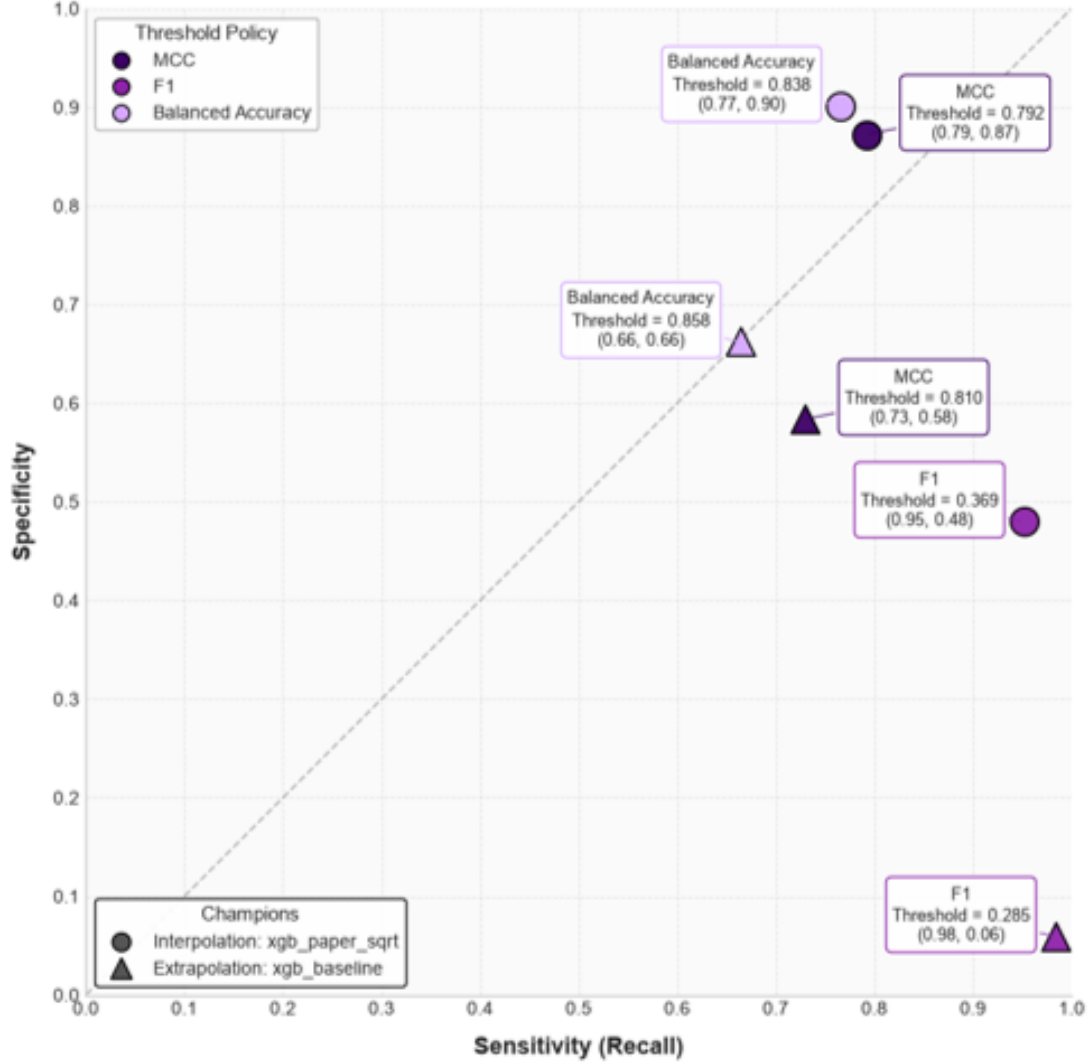


Figure 5.1: Ignition operating points induced by three threshold policies for the interpolation champion (circles) and the extrapolation champion (triangles). Axes are sensitivity (recall) and specificity. F1 selects a low threshold and collapses specificity, especially under paper hold-out ($\tau \approx 0.29$, specificity 0.06). Balanced accuracy and MCC select high thresholds that remain usable in interpolation but only moderately balanced—or worse—once entire papers are withheld.

5.1.5 Model dependence, metrics, and uncertainty

Model rankings are contingent on the tested candidates, search spaces, preprocessing, loss functions, seeds, and selection policy. The configuration files declare 34 FSR and 26 ignition candidates, but the persisted merged FSR manifest contains 25 candidates and the formal ignition selection record omits the available MLP shards. The selected FSR champion was random forest rather than XGBoost, while ignition selected XGBoost. These results apply to the persisted comparison pools; they neither establish complete execution of every configured recipe nor prove that the selected model is optimal among all possible algorithms.

Likewise, a point estimate is not a complete account of performance. R^2 is sensitive to the target variance in a held-out fold and can be unstable when a small set of papers defines a narrow or shifted range. FSR MAPE is extremely large because percentage error is ill-conditioned when observed spread approaches zero; RMSE and MAE in physical units are more interpretable here. For ignition, ROC–AUC measures ranking across thresholds but not probability calibration or a mission-specific operating point. The paper-cluster bootstrap, repeated splits, per-paper metrics, calibration outputs, and fold standard deviations should therefore accompany any headline number.

The final refitted artifacts also have a different evidential status from the outer-fold results. Refitting uses all labelled rows and is appropriate for producing a deployable model, but its training fit is not an unbiased estimate of future performance. Only the nested, untouched outer-fold predictions support comparative claims. The model cards correctly preserve this distinction.

5.1.6 SHAP and the boundary between explanation and causation

The explainability outputs describe model behaviour, not causal mechanisms. For ignition, the global XGBoost SHAP ranking assigns high mean absolute contributions to oxygen, absolute flow speed, Prandtl number, gas molar mass, fuel pyrolysis temperature, and ignition time. Yet grouped permutation importance places fuel pyrolysis temperature, ignition time, fuel heat capacity, thermal diffusion time, and a sample dimension at the top. For FSR, SHAP ranks thermal diffusion time and oxygen prominently, while held-out permutation importance ranks oxygen, flow velocity, gas thermal conductivity, and thermal diffusion time most strongly. Differences are expected because SHAP and permutation importance answer different questions and react differently to correlated inputs.

A SHAP value is an additive decomposition of a particular fitted function relative to a background distribution. It does not show that intervening on that feature would produce the attributed change. Correlated and derived variables allow credit to be redistributed among substitutes. Source confounding can make an apparatus-correlated variable appear physically decisive. Missingness patterns may attract attribution. Mean absolute SHAP also suppresses direction, interactions, and regime dependence. The physically plausible appearance of many rankings is reassuring, but plausibility is not validation of a causal law. Claims such as “oxygen causes the predicted increase” require intervention, mechanistic identification, or a causal design that this observational database does not provide.

The safest interpretation is predictive: within the fitted model and sampled distribution, these features are influential for separating or estimating outcomes. The permutation results add a transfer-oriented check by measuring score degradation on held-out data, but they remain model- and sample-dependent. Explainability is therefore best used to audit whether the learner relies on obviously inappropriate signals, to generate hypotheses, and to identify data-collection priorities. It should not replace combustion theory or controlled experimentation.

5.2 Comparison with the Literature

5.2.1 Rivera/Jose wire-specific machine learning

Rivera, Jose, and colleagues report a literature-derived electrical-wire FSR study in which categorical regime descriptors and augmentation improve random ten-fold cross-validation performance, with several nonlinear models exceeding $R^2 = 0.9$ [7]. Their database contains approximately 1,200 wire observations and explicitly represents core and insulation materials, spread direction, and configuration. Categorisation raises the reported performance of gradient boosting, KNN, SVM, decision trees, and MLP, while bootstrap augmentation is particularly effective for gradient boosting and KNN. This supports two conclusions also visible here: flame spread is strongly nonlinear, and categorical information can expose regime structure that numeric properties alone may not capture.

The numerical results must not be presented as if they were generated by equivalent protocols. The Rivera/Jose work uses random cross-validation and explicitly characterises its result as robustness within the compiled database rather than strict leave-source-out transfer. Its scope is electrical wires, and its target is reported in mm s^{-1} . The present FSR database is broader in geometry and source coverage, reports the model target in ms^{-1} , and makes paper-disjoint nested evaluation the primary scientific test. Hyperparameter search, feature definitions, data versions, augmentation, and source composition also differ. A direct statement that 0.458 is “worse than” > 0.9 would confound task and protocol.

The fair comparison is question-specific. On row-level interpolation, this study’s random-forest $R^2 = 0.879 \pm 0.018$ is close to, though generally below, the strongest wire-specific random-CV values. On paper-level extrapolation, the same family falls to $R^2 = 0.458 \pm 0.222$. Rivera/Jose do not report an equivalent paper-disjoint score, so their > 0.9 result neither contradicts nor establishes performance on this harder transfer task. Instead, their stated caveat and the present protocol gap jointly reinforce the need to name the validation unit whenever quoting accuracy.

There is also a substantive difference in what categorisation can encode. Wire-specific labels such as insulation chemistry, core material, concurrent/opposed spread, and orientation can partition physically distinct regimes. In a broader database, adding similarly detailed categories may improve prediction, but it may also increase sparsity and source fingerprinting. The present models exclude explicit paper identifiers and post-outcome features, yet legitimate categories can remain highly campaign-specific. Their value should therefore be tested under grouped validation, not assumed from random-CV gains.

The augmentation comparison is similarly instructive. Rivera/Jose find bootstrap resampling particularly useful, while noise injection and GAN augmentation can create statistically plausible but not necessarily physically consistent combinations. The present FSR benchmark treats bootstrap copies as reweighting observed conditions; copies inherit the source paper and are never added to the test set. Its selected FSR champions use the real-data condition rather than bootstrap augmentation. Thus, augmentation did not compensate for source shift in the selected model. This supports a conservative principle: resampling can alter optimisation and density, but it cannot supply a new material, apparatus, or gravity regime.

5.2.2 XGBoost and the tabular-learning literature

The strength of tree ensembles is consistent with broader tabular-learning evidence. Shwartz-Ziv and Armon compare XGBoost with several deep tabular architectures on eleven datasets and find that XGBoost generally outperforms individual deep models, requires less tuning, and is less sensitive to whether a dataset appeared in the proposing model’s original paper [9]. The present database has precisely the characteristics that often favour tree ensembles: mixed numeric and categorical predictors, nonlinear interactions, sparse cells, missing fields, unequal scales, and modest sample size relative to the variety of regimes.

The ignition outcome fits this expectation strongly. XGBoost occupies the top of both interpolation and grouped leaderboards; the paper-square-root variant is selected for interpolation and the unweighted baseline for extrapolation. By comparison, decision tree, KNN, and SVM candidates generally rank lower, while the tested focal XGBoost variants perform near chance. This last result is a reminder that a generally strong architecture does not rescue an unsuitable objective implementation or weighting combination.

The FSR result is more nuanced. Random forest narrowly leads interpolation and is selected for grouped transfer, while several XGBoost variants remain competitive. The selected random forest uses no paper weighting, no augmentation, and standard identity-target regression. Its grouped R^2 is 0.458, compared with 0.441 for the strongest listed augmented paper-weighted XGBoost candidate. This difference is small relative to fold variation and should not be elevated into a universal claim that bagging is superior to boosting. It shows only that, within the tested portfolio and policy, the random-forest baseline offered the preferred accuracy–stability combination.

The MLP candidates do not outperform the tree ensembles, particularly under grouped FSR evaluation, where their mean R^2 values can become negative and highly variable. This agrees directionally with the tabular literature, but the explanation is not that neural networks are intrinsically inappropriate. Data volume, regularisation, architecture, optimisation, and representation all matter. Shwartz-Ziv and Armon also find that ensembles combining deep models with XGBoost can outperform either component alone [9]. The present work did not evaluate a formally stacked, outer-fold-safe heterogeneous ensemble, so it cannot rule out such gains.

5.2.3 Physics-informed learning and combustion knowledge

Physics-informed neural networks constrain learning with residuals of governing differential equations and boundary or initial conditions [6]. More broadly, physics-informed machine learning seeks to combine data with conservation laws, symmetries, constitutive knowledge, or admissibility constraints [3]. This is attractive for flame spread because experimental coverage is sparse, predictions must remain physically plausible, and extrapolation is precisely where unconstrained correlations are weakest.

The present models are physics-featured rather than physics-constrained. Inputs include thermophysical properties, dimensionless groups, geometry, oxygen, pressure, flow, gravity, and ignition conditions, but the learned function is not required to satisfy mass, species, momentum, or energy balances. Optional monotonic constraints on oxygen exist for selected XGBoost configurations, yet a monotonic feature constraint is not equivalent to a combustion model. Likewise, categorising gravity or spread direction supplies domain structure but does not create a PINN.

Applying a conventional PINN directly would require an agreed governing formulation, state fields, boundary conditions, and sufficient spatial or temporal observations. This database predominantly contains tabular experimental summaries and endpoint outcomes, not common-domain field solutions. The Rivera/Jose paper also notes that the theoretical framework remains incomplete for direct physics-informed deployment [7]. Therefore, the immediate opportunity is more modest: enforce nonnegative FSR, use dimensionally consistent representations, penalise known inadmissible combinations, model regime-specific monotonicity where defensible, or hybridise reduced-order heat-transfer relations with learned residuals. Such constraints must still be evaluated on held-out sources; physical plausibility alone does not guarantee empirical transfer.

5.2.4 Spacecraft fire-safety context

Guibaud and colleagues identify fire as a major concern for inhabited exploration, particularly as missions move farther from immediate Earth support and involve different atmospheres, gravity levels, confinement, and operational constraints [1]. In that setting, ignition and flame spread are not abstract prediction targets. They contribute to material selection, wire and component design, experiment prioritisation, contingency planning, and the evaluation of how a local ignition might develop.

The present findings add a data-governance qualification to that motivation. Historical evidence can support useful interpolation, but mission-relevant environments may resemble a new source domain rather than another row from a known campaign. Lunar and Martian partial gravity, altered oxygen and pressure, aging materials, realistic ventilation, and integrated spacecraft geometries may lie in sparsely observed combinations. The correct literature-level conclusion is therefore not that machine learning replaces fire testing, but that it can organise heterogeneous evidence and identify where that evidence is or is not transferable.

5.3 Real-World Applications (approximately 30%)

5.3.1 Screening and prioritisation

The most defensible near-term application is screening. Given candidate conditions in the documented schema, the ignition artifact can rank cases by predicted ignition probability, and the FSR artifact can estimate spread rate for records sufficiently similar to the training support. This can reduce a large design space to a smaller set requiring detailed simulation or experiment. A screening tool can, for example, flag material–geometry–atmosphere combinations whose predictions and uncertainty suggest elevated concern, or prioritise boundary cases where models disagree.

Such use should be explicitly asymmetric. A high-risk prediction may justify additional investigation; a low-risk prediction should not by itself waive testing. False reassurance is particularly consequential in spacecraft fire safety. Threshold selection must therefore follow an operational loss function rather than a generic metric. If missing a potentially ignitable case is costly, a sensitivity-oriented threshold may be appropriate, accepting more false alarms. If scarce test capacity is the binding constraint, a more specific threshold may be chosen, but the resulting missed-case risk must be quantified. Four policies—MCC, F1, balanced accuracy, and Youden J —are retained rather than declaring one universally correct; figure 5.1 shows that F1 in particular induces an almost-always-positive rule under paper hold-out.

FSR screening should rely primarily on RMSE and MAE in physical units, not MAPE near zero. Predictions should be accompanied by a domain flag and an interval or empirical error summary. The grouped champion’s mean RMSE of 0.0122 m s^{-1} , with substantial fold variation, is large relative to many low-spread observations. Ranking may therefore be more reliable than precise point estimation in underrepresented regimes. Where a decision depends on a narrow FSR boundary, confirmatory testing is necessary.

5.3.2 Experiment design and the unexplored oxygen–pressure domain

A second application is directing new data collection. Per-paper errors can identify campaigns that are systematically difficult to predict. Disagreement between random and grouped performance identifies fields that may encode source-specific rather than transferable information. Explainability, used cautiously, can nominate influential variables whose measurement completeness deserves improvement. For example, the repeated importance of fuel pyrolysis temperature, ignition time, thermal diffusion time, oxygen, flow, and gas properties suggests that missing or inconsistently defined values in these fields can be costly.

The most direct map of what remains to be tested is the oxygen–pressure coverage of the assembled table (figure 5.2). Panel (a) plots every observation in oxygen mole fraction versus pressure, coloured by ignition outcome. Panel (b) bins the same plane by sampling density; empty cells are empirical gaps, not physically forbidden regions. Most rows cluster in a horizontal band near terrestrial pressure ($\approx 100 \text{ kPa}$) and in vertical stripes at a few laboratory oxygen set-points (0.2, 0.3, 0.4, 0.5, and 1.0). Ignition and no-ignition outcomes overlap inside that band: neither oxygen nor pressure alone separates the label, which is why a univariate oxygen cutoff is not a substitute for the multivariate model. Equally important are the white regions. Oxygen mole fractions between about 0.55 and 0.95 are almost unsampled except at isolated pressures; low oxygen below 0.1 is nearly empty; high pressure above a few megapascals is visited only at a handful of oxygen fractions; and low pressure below $\approx 20 \text{ kPa}$ is sparse away from a small low-oxygen tail. Proposed exploration atmospheres—reduced total pressure with elevated oxygen fraction—sit precisely in combinations that the density plot leaves blank or thinly populated.

This geometry is the practical content of “out of distribution” for a spacecraft material screener. A query whose oxygen and pressure each lie inside the marginal ranges can still fall in an empty joint cell. Predictions there are extrapolations in the experimental design sense, even if the model API returns a number. The figure therefore does not merely decorate the database: it is a search-space map. Future tests should be allocated to those empty oxygen–pressure cells, crossed with underrepresented geometries, materials, and gravity platforms, rather than added as further dense repeats of the 100 kPa band. Because spaceflight experiments are expensive, model

uncertainty and this coverage plot can help allocate them, but the acquisition strategy should reserve samples for prospective validation rather than using every new result for immediate retraining.

A proposed test condition can be compared with ranges, categories, and combinations represented in training. A case in a familiar oxygen–pressure range but a new material–geometry–gravity combination should still be labelled out of distribution. Partial-gravity testing remains an especially strong candidate for prioritisation: lunar- or Martian-relevant gravity, with controlled atmosphere and flow, would test whether a numeric gravity feature transfers smoothly or whether regime-specific dynamics dominate.

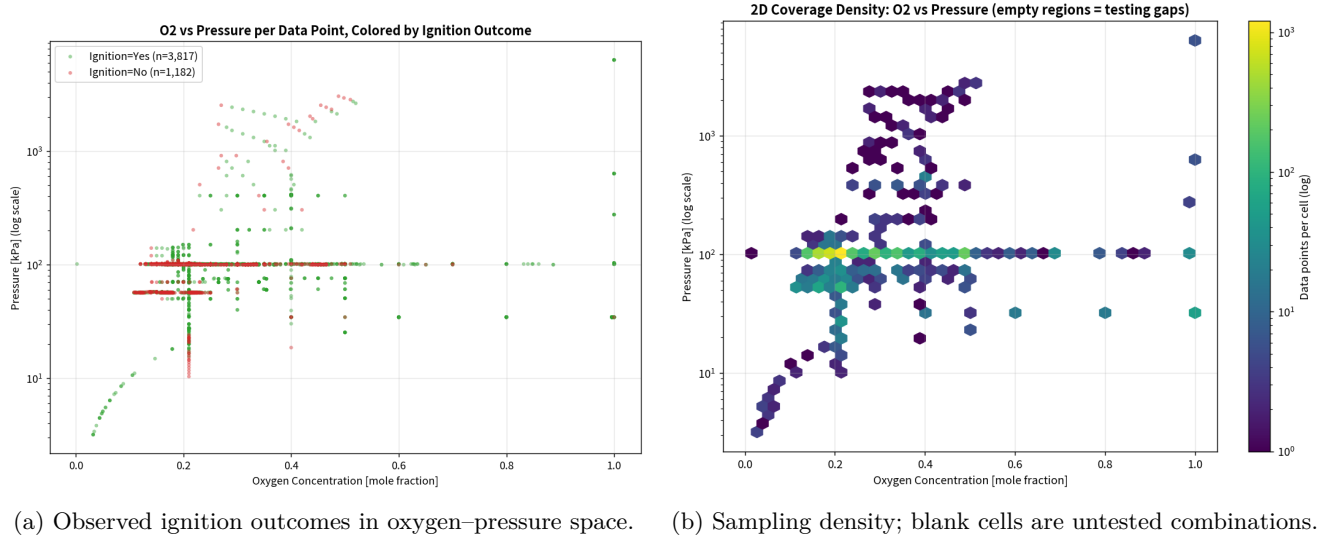


Figure 5.2: Oxygen mole fraction versus pressure in the assembled database (logarithmic pressure axis). The dense terrestrial-pressure band and a few discrete oxygen set-points dominate the literature. Empty regions—high oxygen at intermediate pressure, very low oxygen, and much of the high-pressure and low-pressure envelopes—are the experimental search space that screening still cannot claim.

5.3.3 Material and design decision support

Within a formal engineering workflow, predictions can be one line of evidence in a multi-criteria decision analysis. Candidate materials may be evaluated on ignition propensity and expected spread alongside toxicity, smoke, mechanical performance, electrical function, mass, manufacturability, aging, and compatibility. The models can reveal trade-offs: one material may score lower for ignition under represented conditions yet have uncertain spread behaviour in a new geometry. Presenting both tasks separately is essential because ignition probability and post-ignition spread rate are related but not interchangeable safety endpoints.

A suitable report for each candidate would contain: the input record and units; missing or imputed fields; the model and data fingerprint; interpolation and grouped evidence; the selected threshold policy; predicted probability or FSR; uncertainty or fold-based error context; nearest supported regimes; out-of-distribution warnings; and known limitations. It should also record whether the recommendation is merely “test first”, “requires expert review”, or “within represented screening envelope”. Raw probabilities without this provenance invite overinterpretation.

Human review remains central. A combustion specialist should assess whether the implied trend is physically credible and whether the case violates assumptions hidden by the tabular representation. A materials or systems engineer should evaluate integration effects absent from the database. A safety authority should determine the acceptable evidence and margins. Model output is an aid to these roles, not an automated disposition.

5.3.4 Uncertainty, abstention, and out-of-distribution use

The current repeated-fold standard deviations and cluster-bootstrap intervals provide evaluation uncertainty, not a complete case-specific predictive distribution. Operational use needs both aleatoric uncertainty, arising from experimental variability, and epistemic uncertainty, arising from sparse support and model uncertainty. An ensemble across seeds, folds, or model families could estimate disagreement. Conformal methods could provide empirical intervals or prediction sets if calibrated under a defensible exchangeability assumption. Neither removes distribution shift; grouped or source-conditional calibration remains necessary.

Abstention is a safety feature. The system should decline to issue a routine prediction when required fields are absent, categorical levels are unseen, the query is distant from training support, model portfolios disagree substantially, or a physically admissible range is violated. Out-of-distribution detection should consider mixed data types and joint structure. A simple min–max range check is insufficient because impossible or unsupported combinations can lie inside all marginal ranges.

For ignition, calibration must be examined under held-out papers. A nominal probability of 0.8 is operationally meaningful only if similar grouped predictions ignite at approximately that rate, within uncertainty. For FSR, an interval should be checked for coverage by paper and regime, not only pooled rows. Performance dashboards should expose the worst papers and regimes rather than average away failure pockets.

5.3.5 No replacement for certification or qualification

These artifacts are not certification models. They have not undergone prospective external validation, formal software qualification, independent verification and validation, uncertainty qualification for a specified design envelope, or acceptance by a regulatory or mission assurance authority. The database is observational and heterogeneous; source-held-out performance is moderate; and the models do not enforce governing physics. No model output should replace required material flammability tests, wire qualification, hazard analysis, engineering judgement, or certification evidence.

This limitation does not make the models unusable. Screening and decision support can be valuable precisely because they precede authoritative testing. The proper claim is that the models summarise existing evidence and can prioritise resources under documented uncertainty. The improper claim is that a predicted “no ignition” or low FSR establishes compliance.

Deployment would require version control and auditability. Data hashes, feature manifests, split assignments, selected hyperparameters, model cards, and out-of-fold predictions are already recorded. An operational service should additionally lock the software environment, validate units and schemas, log every query and response, monitor drift, and trigger review when the database or artifact changes. Evaluation artifacts and refitted artifacts must remain clearly separated: the former support performance claims; the latter generate predictions.

5.3.6 A staged deployment concept

A cautious implementation can proceed in stages. In an offline research stage, users inspect leaderboards, paper-level errors, learning curves, calibration, and explanations without affecting decisions. In a shadow stage, predictions accompany conventional assessments but are not used to change them; discrepancies are adjudicated and logged. In a controlled screening stage, outputs may prioritise tests within a predeclared envelope, with mandatory abstention rules and expert sign-off. Only after external prospective evidence could a broader role be considered.

Success criteria should be decision-centred. A useful screener is not merely one with a high ROC-AUC; it reduces testing burden or identifies hazardous cases without exceeding a tolerated miss rate, while remaining calibrated across sources. A useful regressor does not merely minimise pooled RMSE; it provides intervals that cover new-paper observations and rankings that support experimental prioritisation. These criteria align model development with fire-safety outcomes rather than leaderboard optimisation.

5.4 Future Work (approximately 10%)

The immediate priority is external, prospective validation. New experiments should be registered before outcomes are observed, processed through a frozen artifact, and retained as an untouched external test set. Validation should report both pooled and source-specific performance, calibration, abstention frequency, and errors by material, geometry, atmosphere, and gravity regime. Only after this evaluation should the new records enter training.

More partial-gravity data are needed. Purposeful lunar- and Martian-gravity campaigns should vary oxygen, pressure, flow, material, and geometry sufficiently to distinguish gravity effects from source and apparatus effects. Replicate measurements and harmonised uncertainty reporting would separate experimental variability from domain shift.

Uncertainty and out-of-distribution detection should become first-class outputs. Paper-aware ensembles, grouped conformal calibration where assumptions are supportable, mixed-type distance or density models, and explicit abstention thresholds should be evaluated prospectively. Coverage should be measured by source and regime, not only globally.

The database would benefit from a shared ontology: canonical material identifiers; geometry and orientation definitions; units; uncertainty fields; missingness reasons; facility, rig, and diagnostic metadata; and controlled vocabularies for ignition and extinction outcomes. Better provenance would allow laboratory- or facility-held-out validation and reduce accidental source fingerprints.

Finally, physically constrained models should be tested without assuming that a full PINN is immediately appropriate. Useful steps include nonnegative FSR outputs, dimensionally consistent features, regime-aware monotonic constraints, conservation-inspired penalties, and hybrids that learn corrections to reduced-order flame-spread relations. Fully physics-informed formulations become credible only when governing equations, boundary conditions, and compatible field data are available [3, 6]. Every proposed constraint must be assessed under the same paper-disjoint and prospective protocols used to test statistical models.

6 Conclusion

This research asked whether heterogeneous literature data can support reliable machine-learning prediction of material ignition and flame spread rate for fire-safe spacecraft design, especially when prediction is required for an experimental source that the model has not previously seen. The precise answer is conditional. The database supports strong interpolation among rows from represented papers, but only moderate and substantially less stable generalisation to held-out papers. It is therefore suitable as the basis of a traceable screening and research-prioritisation tool, but it does not yet support universal prediction, causal claims, or replacement of qualification and certification testing.

For FSR regression, 2,531 usable observations from 66 canonical papers were evaluated with frozen random-row and paper-grouped holdouts. The selected random-forest baseline achieved $R^2 = 0.879 \pm 0.018$, RMSE = $0.00626 \pm 0.00047 \text{ m s}^{-1}$, and MAE = $0.00210 \pm 0.00017 \text{ m s}^{-1}$ under random-row interpolation. When complete papers were withheld, the same model achieved $R^2 = 0.458 \pm 0.222$, RMSE = $0.01221 \pm 0.00412 \text{ m s}^{-1}$, and MAE = $0.00482 \pm 0.00114 \text{ m s}^{-1}$. For ignition classification, 4,487 deduplicated observations from 87 papers were evaluated. The paper-square-root-weighted XGBoost interpolation champion achieved ROC-AUC = 0.908 ± 0.010 and PR-AUC = 0.969 ± 0.004 ; the unweighted XGBoost grouped champion achieved ROC-AUC = 0.718 ± 0.044 , PR-AUC = 0.877 ± 0.030 , and Brier score = 0.179 ± 0.017 .

The gap between these protocols is the primary result. Random cross-validation answers whether a model can recover nearby conditions in campaigns it has partly observed. Paper-grouped validation asks whether it can transfer across a publication boundary that bundles apparatus, laboratory practice, sampled regime, and reporting convention. The latter is the more relevant internal evidence for a new campaign, although it is not a substitute for external validation. Quoting only the random-split scores would materially overstate readiness.

The model comparison also answers the algorithmic part of the question. Tree ensembles are well matched to these mixed, sparse, nonlinear tabular data. XGBoost led ignition prediction, while random forest was selected for FSR; KNN, SVM/SVR, decision-tree, and MLP candidates supplied useful baselines but did not displace the grouped champions. This is consistent with tabular-learning literature, while also demonstrating that XGBoost is not universally best [9]. The wire-specific $R^2 > 0.9$ findings of Rivera/Jose and colleagues remain compatible with the present results because they concern random cross-validation within a narrower electrical-wire database, not paper-disjoint transfer [7].

Feature attributions identified physically plausible variables, including oxygen, flow, thermal diffusion time, fuel pyrolysis temperature, ignition time, and gas or material properties. These findings are diagnostic rather than causal. SHAP decomposes the fitted model; it does not identify the effect of an intervention. Correlated derived variables, uneven missingness, and campaign confounding prevent causal interpretation.

Accordingly, the practical contribution is a reproducible decision-support framework. It freezes splits, isolates tuning within outer training data, verifies paper disjointness, records data fingerprints and feature manifests, reports repeated-fold and paper-level uncertainty, and distinguishes unbiased evaluation from full-data refitting. Within a documented applicability domain, its predictions can rank candidate tests, identify poorly covered regimes, and focus expert attention. They must be accompanied by uncertainty, out-of-distribution checks, threshold rationale, and explicit abstention.

The work therefore establishes a useful but bounded capability: literature-scale machine learning can extract transferable signal for spacecraft fire-safety screening, yet database heterogeneity and source shift remain the dominant obstacles. The path to stronger evidence is not a larger headline score on mixed rows. It is prospective external validation, targeted partial-gravity experiments, a harmonised ontology, uncertainty-aware abstention, and physically constrained models evaluated against genuinely new sources. Until those steps are completed, the system should inform experiments and engineering judgement, never replace them.

A Technical Appendices

A.1 Feature dictionary

The deployed manifests are the authoritative machine-readable feature definitions. Tables A.1 and A.2 provide a compact human-readable dictionary. Units are encoded in names where available; otherwise they follow the database schema. Derived quantities are predictors, not independent measurements.

Table A.1: Ignition feature dictionary (**table-data-v5**; 32 numeric and five categorical features).

Group	Features and interpretation
Fuel properties	<code>fuel_density_kg_m3</code> , <code>fuel_k_w_mk</code> , <code>fuel_cp_j_kgk</code> , <code>fuel_pyrolysis_t_k</code> , <code>fuel_alpha_m2_s</code> , and <code>fuel_volumetric_heat_capacity_j_m3k</code> : density, conductivity, specific heat, pyrolysis temperature, diffusivity, and volumetric heat capacity.
Core properties	<code>core_density_kg_m3</code> , <code>core_k_w_mk</code> , <code>core_cp_j_kgk</code> , and <code>core_volumetric_heat_capacity_j_m3k</code> .
Gas and transport	<code>oxygen</code> , <code>gas_molar_mass</code> , <code>gas_cp_j_kgk</code> , <code>gas_k_w_mk</code> , <code>gas_density_kg_m3</code> , <code>gas_alpha_m2_s</code> , <code>gas_nu_m2_s</code> , <code>reynolds</code> , <code>peclet</code> , <code>prandtl</code> , and <code>thermal_diffusion_time_s</code> .
Operating conditions	<code>pressure_pa</code> , <code>signed_flow_velocity_m_s</code> , <code>flow_speed_abs_m_s</code> , <code>gravity_g</code> , <code>ignition_power_w</code> , and <code>ignition_time_s</code> .
Geometry	<code>half_thickness_m</code> , <code>characteristic_length_m</code> , and <code>sample_dim_1_m</code> through <code>sample_dim_3_m</code> .
Categorical regime	<code>geometry_cat</code> (wire, flat, cylindrical, spherical), <code>diluent_cat</code> , <code>flow_direction_cat</code> (coflow, counterflow, quiescent), <code>gravity_regime_cat</code> (microgravity, partial, Earth, hyper/unknown), and <code>ignition_method_cat</code> .

Table A.2: FSR feature dictionary (**fsr-data-v2**; exact availability follows the artifact manifest).

Group	Manifest fields
Geometry	Sample dimensions, <code>half_thickness</code> , <code>characteristic_length_m</code> , and categorical <code>Geometry of Sample</code> .
Fuel properties	<code>fuel_density_kg_m3</code> , <code>fuel_k_W_mK</code> , <code>fuel_cp_J_kgK</code> , <code>fuel_pyrolysis_T_K</code> , <code>fuel_alpha_m2_s</code> , and <code>fuel_volumetric_heat_capacity_J_m3K</code> .
Core properties	<code>core_density_kg_m3</code> , <code>core_k_W_mK</code> , <code>core_cp_J_kgK</code> , and <code>core_volumetric_heat_capacity_J_m3K</code> .
Gas and transport	Oxygen Concentration, <code>gas_M</code> , <code>gas_cp_mass</code> , <code>gas_k</code> , <code>gas_density_kg_m3</code> , <code>gas_alpha_m2_s</code> , <code>gas_nu_m2_s</code> , <code>Reynolds</code> , <code>Peclet</code> , <code>Prandtl</code> , and <code>thermal_diffusion_time</code> .
Conditions and apparatus	Pressure (Pa), Flow Velocity (m/s), Gravity, Ignition Power (W), Ignition Time (s), categorical diluent, Ignition Method, and antecedent categorical <code>Ignition</code> .

Explicit source identifiers, DOI/citation fields, duplicate FSR targets, and post-outcome variables such as flame length, heat-release rate, smoke, and extinction indicators are excluded from predictors. Paper identity is retained only for grouping and weighting. Numeric parsing accepts cells beginning with a number; strings such as N2 and CO2 remain categorical. For non-XGBoost models, numeric missing values are median-imputed and categorical values receive a constant unknown level; XGBoost may retain native numeric missingness.

A.2 Metric definitions

For observed FSR y_i , prediction $\hat{y}_i$, sample mean $\bar{y}$, and n test observations,

$$R^2 = 1 - \frac{\sum_{i=1}^n (y_i - \hat{y}_i)^2}{\sum_{i=1}^n (y_i - \bar{y})^2}, \quad (\text{A.1})$$

$$\text{RMSE} = \sqrt{\frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2}, \quad (\text{A.2})$$

$$\text{MAE} = \frac{1}{n} \sum_{i=1}^n |y_i - \hat{y}_i|, \quad (\text{A.3})$$

$$\text{MBE} = \frac{1}{n} \sum_{i=1}^n (\hat{y}_i - y_i), \quad (\text{A.4})$$

$$\text{MAPE} = \frac{100}{n} \sum_{i=1}^n \left| \frac{y_i - \hat{y}_i}{y_i} \right|. \quad (\text{A.5})$$

NRMSE is a normalised RMSE. MAPE is unstable when y_i approaches zero and is secondary to RMSE and MAE for this FSR distribution.

For ignition probability $\hat{p}_i$, the Brier score is

$$\text{Brier} = \frac{1}{n} \sum_{i=1}^n (\hat{p}_i - y_i)^2. \quad (\text{A.6})$$

ROC–AUC measures ranking of positives above negatives across thresholds. PR–AUC summarises precision–recall performance and must be read in light of class prevalence. Given threshold τ ,

$$\hat{y}_i = \mathbf{1}[\hat{p}_i \geq \tau]. \quad (\text{A.7})$$

Sensitivity is $\text{TP}/(\text{TP} + \text{FN})$, specificity is $\text{TN}/(\text{TN} + \text{FP})$, precision is $\text{TP}/(\text{TP} + \text{FP})$, and balanced accuracy is the mean of sensitivity and specificity. The F1 score is the harmonic mean

$$\text{F1} = \frac{2 \text{precision} \cdot \text{recall}}{\text{precision} + \text{recall}}. \quad (\text{A.8})$$

The Matthews correlation coefficient is

$$\text{MCC} = \frac{\text{TPTN} - \text{FPFN}}{\sqrt{(\text{TP} + \text{FP})(\text{TP} + \text{FN})(\text{TN} + \text{FP})(\text{TN} + \text{FN})}}. \quad (\text{A.9})$$

Thresholds optimising MCC, F1, balanced accuracy, and Youden’s J are selected from inner out-of-fold predictions, never from the corresponding outer test labels.

A.3 Reproducibility workflow

The reproducibility contract is:

1. Load the named database and record its SHA-256 fingerprint, data version, parsing decisions, target counts, missingness, and duplicate removal.
2. Canonicalise DOI or citation into `paper_id`; exclude source identifiers and post-outcome leakage from model inputs.
3. Freeze all row and paper-grouped split assignments before model comparison. Assert complete row coverage, no row overlap, and zero paper overlap for grouped protocols.
4. For every candidate and outer fold, fit preprocessing only on the outer training partition. Tune hyperparameters on inner folds; use group-disjoint inner folds when the outer protocol is grouped.
5. Fit the selected inner configuration on the complete outer training partition and score the untouched outer test partition once. For ignition, freeze thresholds from inner out-of-fold probabilities. For FSR bootstrap conditions, augment training only and preserve inherited paper identity.
6. Persist outer predictions, fold and per-paper metrics, search histories, selected hyperparameters, integrity checks, and failure records. Aggregate only complete, valid candidates.
7. Select champions from persisted tables under the versioned policy. The FSR policy prioritises RMSE; the ignition policy prioritises ROC–AUC and requires LOPO robustness for extrapolation.

8. Refit the selected configuration on all labelled data for inference, package the model card and feature manifest, and retain evaluation results separately. A refitted artifact is deployable; outer-fold predictions are the unbiased comparison evidence.

The recorded fingerprints are shown below; random seeds begin at 42 and split files are shared across candidates.

FSR: 6e09a99e4ea69148e69151bd46149be4140d181b2b8294106927ea9499002fc8
 Ignition: d80da0648ca3a7d77d31ae90e53b4e81b2ab28c8ec666ce6cd150ee7544f6

A.4 Traceability matrix

Table A.3: Claim-to-evidence traceability matrix.

Claim	Persisted evidence
FSR dataset has 2,531 target rows from 66 papers	Dataset-integrity summary for the FSR modeling cohort.
FSR interpolation and grouped champion scores	Frozen interpolation and extrapolation leaderboards and the FSR selection record.
FSR transfer loss and stability	Repeated-holdout gap and metric-stability tables.
Ignition dataset has 4,487 rows, 87 papers, and 3,383/1,104 class counts	Dataset-integrity summary for the ignition modeling cohort.
Ignition interpolation and grouped scores	Frozen interpolation and extrapolation leaderboards.
Threshold trade-offs	Champion threshold-performance table used in figure 5.1.
Explanations are model-dependent	SHAP and permutation-importance tables for each champion.
Champion identity and full-data artifact provenance	Selection records, model cards, feature manifests, and training-data fingerprints.
No paper leakage and nested tuning contract	Split-assignment checks and evaluation integrity records.

A.5 Oral-defense reconciliation

The following concise answers reconcile statements that can otherwise appear contradictory.

- Why can the model be called strong if grouped performance is moderate?** It is strong for interpolation and useful but limited for transfer. The report does not collapse those questions into one score.
- Why are Rivera/Jose results above 0.9 while this grouped FSR R^2 is 0.458?** Their headline results concern a narrower wire database under random ten-fold cross-validation; this study’s 0.458 holds out entire papers. The protocols, source composition, features, units, and algorithms are not equivalent [7].
- Does paper-grouped validation prove performance on a future spacecraft?** No. It is stronger internal evidence against paper leakage, but future conditions may differ by laboratory, material, facility, gravity regime, and mission environment. Prospective external validation remains necessary.
- Why use XGBoost if random forest wins FSR?** XGBoost is a strong tabular prior and wins ignition, consistent with the literature [9]; model selection is empirical and task-specific. Random forest offered the preferred grouped FSR result in the tested portfolio.
- Does SHAP identify combustion mechanisms?** No. SHAP explains the fitted prediction relative to a background distribution. Correlation, derived features, missingness, and source confounding preclude causal interpretation.
- Did bootstrap augmentation create new experiments?** No. It resampled observed training rows, inherited their paper IDs, and changed weighting. It added no new physical support.
- Can a low predicted risk certify a material?** No. The artifacts support screening and prioritisation only. Qualification, certification, and expert hazard assessment remain authoritative.
- Why not deploy a PINN immediately?** The database mainly contains heterogeneous tabular endpoints, not common-domain fields with agreed governing equations and boundary conditions. Near-term physically constrained or hybrid models are more defensible [6].

A.6 Model portfolios

The executable YAML defines the configured portfolio: 34 FSR candidates (10 XGBoost, seven random forest, six KNN, five MLP, and six SVR) and 26 ignition candidates. The persisted evidence is narrower: the merged FSR manifest contains 25 candidates, each with real-data (**rd**) and bootstrap-augmented (**rd_bt**) conditions, while the formal ignition selection record omits MLP despite available shard metrics. The selected champions

Table A.4: Configured model portfolios.

Task	Family	Candidate hypotheses
FSR	XGBoost (10)	Baseline; square-root/inverse paper weights; absolute-error and pseudo-Huber objectives; signed-log($1 + y$) target; paper bagging.
FSR	Random forest (7)	Baseline and paper weights; absolute-error criteria; signed-log target variants.
FSR	KNN (6)	Uniform or distance neighbours, each with no, square-root, or inverse paper weighting.
FSR	MLP (5)	Baseline; paper weighting; signed-log target variants.
FSR	SVR (6)	Linear or RBF kernels, each with no, square-root, or inverse paper weighting.
Ignition	XGBoost (8)	Baseline; paper-only and class-only weights; combined square-root/inverse weights; focal-loss variants.
Ignition	Decision tree (3)	Baseline and combined class plus square-root/inverse paper weighting.
Ignition	KNN (5)	Baseline; weighted variants; distance-neighbour variants.
Ignition	MLP (5)	Baseline; weighted variants; focal-loss variants.
Ignition	SVM (5)	RBF baseline/weighted variants and linear-kernel weighted variants.

are `rf_baseline [rd]` for both FSR protocols, `xgb_paper_sqrt` for ignition interpolation, and `xgb_baseline` for ignition extrapolation. These names identify frozen configurations within the persisted comparison pools and do not imply universal superiority outside the recorded schema and validation design.

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